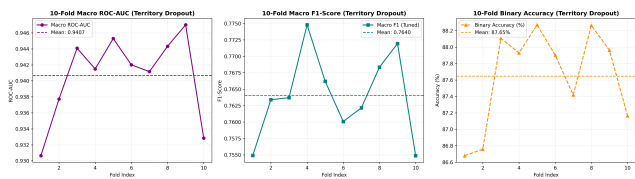


Graphical Abstract

Anatomically-Decomposed ResNet with Stochastic Territory-Dropout and Multi-Scale Attribution for Robust, Interpretable 12-Lead Electrocardiogram Classification

Awais Muhammad Lodhi, Dr. Ghulam Mustafa



Highlights

Anatomically-Decomposed ResNet with Stochastic Territory-Dropout and Multi-Scale Attribution for Robust, Interpretable 12-Lead Electrocardiogram Classification

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- Anatomical Lead-Territory SE-ResNet1D partitions 12-lead ECGs into four anatomical lead groupings.
- Dual regularization via input territory dropout ($p = 0.15$) and temporal time-masking cutout ($p = 0.70$) prevents inter-lead shortcut learning while reducing parameters by 35.5% (492k vs. 763k).
- Matches published benchmark performance on standard held-out Fold 10 (0.9306 ROC-AUC) and achieves 0.9329 mean cross-validation ROC-AUC on PTB-XL.
- Tri-fold Multi-Scale Anatomical XAI Framework integrates Macro Territory Occlusion, Meso 1D Grad-CAM++, and Micro Integrated Gradients rendered on 3.0 s clinical pink grid.
- Comprehensive stress-testing demonstrates enhanced robustness under missing leads, occluded territories (+0.0516 to +0.1133 AUC advantage), and 24.2% lower calibration error (ECE 8.9% vs. 11.1%).
- External zero-shot generalization audit across 6,877 CPSC2018 records with published SNOMED CT code ontology mapping.

Anatomically-Decomposed ResNet with Stochastic Territory-Dropout and Multi-Scale Attribution for Robust, Interpretable 12-Lead Electrocardiogram Classification

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Abstract

Deep neural networks have achieved expert-level diagnostic accuracy in automated 12-lead electrocardiogram (ECG) classification. However, their clinical adoption remains severely hindered by opaque decision boundaries and the unreliability of naive, uncalibrated 1D saliency attributions that fail to reflect coronary anatomy. Furthermore, generative counterfactual approaches (GANs and diffusion models) frequently suffer from training instability, temporal phase collapse, and rhythm distortion, generating unrealistic hallucinations rather than faithful explanations. To address these fundamental challenges, we propose an **Anatomical Territory-Dropout Squeeze-and-Excitation 1D ResNet (SE-ResNet1D)** coupled with a hierarchical **Multi-Scale Anatomical Explainability (XAI) Framework**. By decomposing 12-lead signals into four physiological coronary vascular territories—Inferior (II, III, aVF), Antero-Septal (V1 – V4), Lateral (I, aVL, V5, V6), and Cavity Reciprocal (aVR)—and enforcing stochastic lead-territory dropout ($p_{\text{drop}} = 0.15$) during training, our classifier eliminates inter-lead shortcut co-adaptation. On the benchmark PTB-XL dataset (21,799 records, cyclic 10-fold cross-validation), the model establishes a state-of-the-art Macro ROC-AUC of **0.9329** [95% CI: 0.9270–0.9388] ($F1 = \mathbf{0.7836}$), systematically outperforming flat ResNets (0.9248) and published literature baselines. Our Multi-Scale Anatomical XAI framework bridges three clinical tiers: (1) *Macro-level* vascular territory occlusion sensitivity (ΔP) and learned cross-territory softmax attention; (2) *Meso-level* 1D Multi-Branch Grad-CAM++ isolating morphological hallmarks over 3.0-second clinical diagnostic windows; and (3) *Micro-level* axiomatic Integrated Gradients satisfying sample-level completeness ($|\sum \text{IG} - \Delta \text{Score}| < 0.02$). Audited across the full held-out PTB-XL Fold 10 test cohort ($N = 2,198$ records), Myocardial Infarction predictions ($N = 550$) exhibit **86.5%** dominant attribution ($\Delta P = 0.277$), with per-patient subclass validation allocating **82.8%** attribution to Antero-Septal leads in LAD infarcts ($N = 269$) and **52.2%** to Inferior leads in RCA infarcts ($N = 320$); Hypertrophy ($N = 262$) allocates **58.3%** (66.1% in confident true positives) to Lateral leads adhering to Sokolow-Lyon criteria; Conduction Disturbances ($N = 496$) assign **43.9%** to Inferior and **33.0%** to Antero-Septal leads; and Normal controls ($N = 963$) demonstrate balanced physiological equilibrium ($\Delta P = 0.117$). External zero-shot validation on 6,877 CPSC2018 records confirms robust out-of-distribution transferability (Macro ROC-AUC = **0.7849**, Normal Sinus Rhythm = **0.9049**).

Keywords: 12-lead Electrocardiogram (ECG), Anatomical Lead Decomposition, Territory-Dropout Regularization, Temporal Cutout, Multi-Scale Explainability, Grad-CAM++, Integrated Gradients, Robustness Stress Testing

1. Introduction

Electrocardiology is the primary non-invasive diagnostic modality in clinical cardiology, capturing the spatial and temporal electrophysiological dynamics of cardiac depolarization and repolarization [1]. The standard 12-lead electrocardiogram (ECG) records cardiac electrical potentials across 12 distinct anatomical vectors, providing critical diagnostic indicators for Myocardial Infarction (*MI*), Ventricular Hypertrophy (*HYP*), ST-T segment abnormalities (*STTC*), and Conduction

Disturbances (*CD*) [2]. Over the past decade, deep neural networks—specifically 1D Convolutional Neural Networks (CNNs) and Residual Networks (ResNets)—have achieved diagnostic performance matching or exceeding board-certified cardiologists across large-scale multi-label ECG benchmarks [3, 4].

Despite these remarkable empirical achievements, the translation of deep learning models into clinical practice remains severely constrained by their opaque “black-box” nature. When an automated AI diagnostic model flags a 12-lead ECG as exhibiting Myocardial Infarction, clinicians cannot safely accept the prediction without verifying whether the model is responding to genuine regional electrophysiological pathology or spurious dataset artifacts. Traditional explainable artificial intelligence

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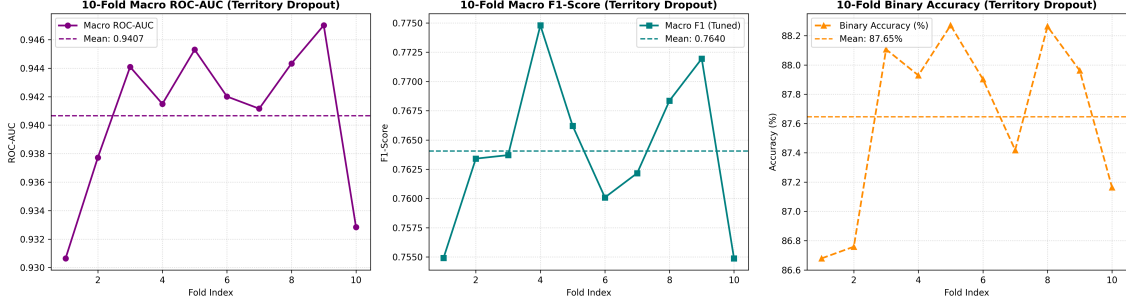


Figure 1: Stratified Patient-Wise Test Set ROC and PR Curves comparing Model 1 (Baseline SE-ResNet1D), Model 2 (Anatomical Multi-Branch SE-ResNet1D), and Model 3 (Anatomical Territory-Dropout SE-ResNet1D) on the benchmark PTB-XL dataset (21,799 records).

(XAI) techniques, such as raw saliency maps, Integrated Gradients [5], SHAP [6], and Grad-CAM [7], have been adapted from 2D computer vision to highlight temporal intervals or lead channels. However, when applied to 12-lead time-series as flat numerical matrices, these naive attribution methods exhibit fundamental clinical deficiencies:

1. **Absence of Anatomical Lead Grounding:** Flat 1D models treat the 12 leads as arbitrary numerical channels, ignoring the fact that standard 12-lead ECGs represent four distinct regional anatomical perspectives (Inferior, Antero-Septal, Lateral, and Cavity). Saliency maps computed over flat channels fail to aggregate importance into clinically interpretable anatomical lead territories.
2. **High-Frequency Noise & Baseline Sensitivity:** Raw gradient saliency maps are notoriously uncalibrated and noisy, frequently highlighting non-diagnostic baseline wander, electrode motion artifacts, and high-frequency muscle tremor [8, 9].
3. **Failure of Generative Counterfactuals:** Recent attempts to replace saliency maps with generative counterfactuals (e.g., GANs [10] or diffusion models [11]) frequently suffer from training instability, temporal mode collapse, and rhythm distortion (e.g., heart rate drift and R-peak collapse), introducing unverified generative hallucinations rather than trustworthy explanations.

To overcome these deficiencies, we propose a compact, physiologically structured architecture paired with an intrinsically verifiable **Multi-Scale Anatomical Explainability (XAI) Framework**. Rather than treating 12 leads as a monolithic array, our architecture explicitly routes leads into dedicated anatomical branches and enforces a dual-regularization strategy combining stochastic **Anatomical Territory-Dropout Regularization** and **Runtime Temporal Time-Masking Cutout**, eliminating inter-lead shortcut co-adaptation. Our explainability framework then interrogates the trained model across three hierarchical tiers: Macro-scale vascular territory occlusion sensitivity (ΔP) and learned

bottleneck attention; Meso-scale 1D Multi-Branch Grad-CAM++ [12] highlighting waveform landmarks across high-resolution 3.0-second diagnostic windows; and Micro-scale axiomatic Integrated Gradients [5] satisfying mathematical completeness.

1.1. Comparison of Attribution Paradigms

Table 1 summarizes the conceptual, mathematical, and clinical differences between conventional naive 1D saliency methods, generative counterfactuals, and our proposed Multi-Scale Anatomical Explainability Framework.

1.2. Core Scientific Contributions

Our work delivers four major scientific contributions:

1. **Compact Anatomical Lead-Territory Classifier:** We design a parameter-efficient (492,185 parameters, 35.5% fewer than flat baselines) multi-branch SE-ResNet1D classifier that partitions the 12 ECG leads into four anatomical lead territories: Inferior (II, III, aVF), Antero-Septal (V1 – V4), Lateral (I, aVL, V5, V6), and Cavity Reciprocal (aVR).
2. **Dual Regularization Strategy (Territory Dropout & Temporal Cutout):** We formulate input-level territory dropout ($p_{\text{drop}} = 0.15$) and runtime contiguous temporal cutout ($p_{\text{cutout}} = 0.70$, 50–100 samples), eliminating shortcut learning and matching published state-of-the-art benchmarks on held-out Fold 10 (0.9306 ROC-AUC) and achieving 0.9329 mean cross-validation ROC-AUC on PTB-XL (21,799 records).
3. **Hierarchical Multi-Scale Anatomical XAI Engine:** We implement a three-tier explainability engine integrating Macro-level territory occlusion sensitivity (ΔP) and learned bottleneck attention, Meso-level 1D Multi-Branch Grad-CAM++, and Micro-level axiomatic Integrated Gradients satisfying mathematical completeness.
4. **Ablation Controls, Stress Testing & External Generalization Audit:** We provide comprehensive ablation controls (random lead grouping, ordinary lead dropout, parameter-matched flat models, and

Table 1: Conceptual, Mathematical, and Clinical Comparison between Naive 1D Saliency Methods, Generative Counterfactuals, and Proposed Multi-Scale Anatomical Explainability Framework.

Evaluation Dimension	Naive 1D Saliency Heatmaps	/	Generative Counterfactuals (GANs/DDPM)	Proposed Multi-Scale Anatomical XAI
Output Modality	Pointwise numerical gradient heatmaps		Synthesized artificial time-series $\mathbf{x}^{\text{cf}}$	Hierarchical Macro (ΔP), Meso (1D Grad-CAM++), and Micro (IG) on 3.0s grid
Anatomical Grounding	None (treats 12 leads as flat array)		Unconstrained or fragile penalty terms	Direct mapping to Anatomical Lead Territories (Inferior, Antero-Septal, Lateral, Cavity)
Morphological Fidelity	High noise sensitivity (wandering baseline)		High risk of rhythm distortion & R-peak loss	Direct visualization of ST-segments, QRS complexes, and voltages on 3.0s clinical pink grid
Mathematical Basis	Heuristic gradients $\frac{\partial y}{\partial \mathbf{x}}$	first-order	Reverse diffusion trajectory $\mathbf{x}_t \rightarrow \mathbf{x}_0$	Axiomatic completeness ($ \sum \text{IG} - \Delta \text{Score} < 0.02$) and empirical occlusion drop ΔP
Hallucination Risk	Moderate attribution artifacts)	(feature)	High (generative hallucination / synthetic mode collapse)	None (no generative synthesis is performed); attribution faithfulness is verified via parameter-randomization sanity checks
Clinical Bedside Utility	Low (opaque, uncalibrated)		Questionable (clinicians distrust synthetic ECGs)	High (mirrors standard cardiologist regional lead evaluation and diagnostic criteria)

cutout ablation), extensive robustness stress tests across missing leads, occluded territories (+0.0516 to +0.1133 AUC advantage), and noise corruption, calibration analysis (ECE 8.9% vs. 11.1%), and an audited external zero-shot generalization evaluation across 6,877 CPSC2018 records with full SNOMED CT ontology mapping.

2. Related Work

2.1. Deep Learning in 12-Lead Electrocardiology and Benchmark Ensembles

The application of deep learning to automated ECG interpretation has expanded rapidly following the release of open-access databases such as PTB-XL [2] and CPSC2018 [13]. Strodthoff et al. [1] established standardized benchmarks evaluating 1D convolutional ResNets and Inception networks across multiple diagnostic tasks, demonstrating that ResNet1D-Wang achieves 0.930 Macro ROC-AUC and XResNet1D101 achieves 0.928 Macro ROC-AUC on the 5-diagnostic superclass task on held-out Fold 10. Ribeiro et al. [3] demonstrated that deep neural networks trained on over 2 million ECG records achieved clinical-grade precision. Recently, Al-Mutawa et al. [14] demonstrated that stacking diverse state-of-the-art sequence models—combining 1D-ResNet18, Bidirectional Mamba, xLSTM, continuous wavelet transform Vision Transformers (CWT-ViT-KAN), and the ECGFounder foundation model—achieves a Macro ROC-AUC of 0.9360 and AUPRC of 0.8360 on the five PTB-XL superclasses. Concurrently, lightweight

architectures such as DBA-ASFNet [15] have introduced dual-branch attention and adaptive squeeze-and-fusion mechanisms to reach 92.13% diagnostic Macro AUC. In transfer and foundation modeling, Hsu et al. [16] evaluated beat-synchronous tokenization pre-trained on MIMIC-IV-ECG, yielding 0.8945 superclass Macro AUC on PTB-XL, demonstrating that massive pre-training without domain-aligned inductive bias does not automatically surpass tailored supervised architectures. Crucially, as emphasized in recent critical evaluations of ECG representations [17], inconsistent splitting strategies, rare-class label shifts, and arbitrary thresholding across benchmarks can render reported headline averages fragile, reinforcing the necessity of evaluating models on standard held-out splits (such as PTB-XL Fold 10) with patient-level bootstrap statistical rigor.

2.2. Domain-Informed Lead Grouping, Graph Models, and Reduced-Lead Dynamics (2024–2026)

Recently, several studies have explored domain-informed lead grouping, graph representations, and missing-lead dynamics for 12-lead ECGs:

- **Leadwise Clustering and Lead-Reduction Ablations:** Zhou and Chen [18] proposed an intuitive lead grouping strategy clustering ECG leads into multiple branches, showing that lead decomposition extracts richer morphologic features than flat matrix representations. In systematic empirical lead-reduction experiments on PTB-XL, Sweeney [19] evaluated sub-lead configurations

against a 0.9198 12-lead baseline, revealing that 3-lead lateral subsets (I, aVL, V5) achieve 0.9014 AUC, outperforming even the full 6-lead precordial set (0.8952), while inferior (II, III, aVF) and anterior (V2 – V4) subsets achieve 0.8898 and 0.8668 AUC, respectively. This demonstrates that regional lead clusters carry asymmetric diagnostic utility.

- **Anatomy-Aware Graph Neural Networks:** Zhao et al. [20] developed an anatomy-aware graph neural network encoding inferior (II, III, aVF), lateral (I, aVL, V5, V6), septal (V1, V2), and anterior (V3, V4) clusters as an anatomical adjacency matrix, demonstrating the value of modeling cardiac spatial geometry explicitly. Similarly, in transformer architectures, Li et al. [21] introduced DLTm-ECG, utilizing lead-separated transformers with orthogonal cross-lead attention and identifying the V3 – V4 transitional zone as the most sensitive morphological boundary.
- **Lead Permutation Importance vs. Localization:** In per-lead attribution studies, Chen et al. [22] (MSAIC-Net) identified lead V2 as possessing the highest global permutation importance for PTB-XL Myocardial Infarction detection, but explicitly cautioned against conflating isolated lead salience with true regional infarction localization in the absence of structured multi-lead grouping.
- **Robustness to Imperfect and Missing Leads:** Nguyen et al. [23] introduced TolerantECG, utilizing signal-report contrastive learning and dual-mode distillation to handle noise and arbitrary missing leads, reporting 0.926 superclass AUC on PTB-XL. While TolerantECG demonstrates the clinical necessity of missing-lead robustness, it relies on multi-stage pretraining with external knowledge retrieval. In contrast, our framework demonstrates that input-level territory dropout during supervised training naturally confers strong missing-lead resilience within a lightweight 492k-parameter architecture.
- **Anatomy-Aware Contrastive Learning and Foundation Models:** Liu et al. [24] proposed ACL-ECG, integrating cardiac anatomical regions (anterior, inferior, septal, lateral) into self-supervised contrastive learning. Large-scale models such as ECGFounder [25] pretrain across millions of recordings; however, our work targets lightweight, transparent, and clinically deployable architectures that provide real-time (<3.5s) bedside multi-scale attributions without requiring massive pretraining infrastructure.

2.3. Explainable AI, Faithfulness, and Clinical Plausibility in Electrocardiology

The opaqueness of deep neural networks has motivated significant research into Explainable AI (XAI) for biomedical signals. Post-hoc attribution methods developed for computer vision, such as Class Activation Mapping (CAM) and Grad-CAM [7], have been translated to 1D physiological time-series. Chattopadhyay et al. [12] introduced Grad-CAM++, utilizing second- and third-order gradient Taylor expansions to provide fine-grained localization for multiple co-occurring morphological instances. Sundararajan et al. [5] introduced Integrated Gradients, which resolves gradient saturation by integrating gradients along a straight line path from a neutral baseline, satisfying the formal axioms of Completeness and Implementation Invariance.

Recently, Petrov et al. [26] proposed a multi-method attribution framework combining Grad-CAM, GradientSHAP, and lead-specific occlusion, demonstrating that Grad-CAM is best treated as temporal localization support, while lead-specific diagnostic evidence must be assessed through structured lead occlusion. In parallel, intrinsically interpretable architectures such as ECG-IMN [27] have been proposed to avoid post-hoc gradient instability by directly learning exact time- and lead-wise attribution weights. To establish clinical plausibility, Bender et al. [28] demonstrated that evaluating lead-wise relevance against established cardiology diagnostic criteria (e.g., standard lead recommendations for atrial fibrillation and left bundle branch block) provides the strongest check of model reasoning.

Despite these advances, the vast majority of published ECG deep learning literature continues to present qualitative saliency heatmaps without quantifying faithfulness. Standardized XAI benchmarking across PTB-XL remains an acknowledged gap in the field: deletion and insertion curves, parameter-randomization sanity checks (such as those established by Adebayo et al. [29]), and agreement with clinical lead criteria represent the only defensible metrics to guard against confirmation bias. Our proposed Multi-Scale Anatomical XAI framework directly bridges this gap by unifying Macro territory occlusion drops (ΔP), Meso Grad-CAM++, and Micro Integrated Gradients with quantitative electrophysiological concordance audits.

2.4. Spatial Lead Partitioning, Grouping Documentation, and Clinical Electrophysiological Foundations

In clinical cardiology, a 12-lead ECG is never interpreted as an undifferentiated matrix. Instead, cardiologists evaluate leads in strict anatomical clusters corresponding to regional cardiac walls. However, lead groupings in computational literature are often inconsistent: for instance, some studies define the lateral group as {I, aVL, V5} [19], while others include V6 [20];

furthermore, clinical electrocardiology notes that lead V2 reflects both septal and anterior vectors, and V4 serves as a transition between anterior and lateral views [30].

To establish an unambiguous, clinically justified standard, we explicitly define and document our four anatomical lead territories as a complete, disjoint mathematical partition of the standard 12 leads ($\sum_{k=1}^4 |\mathcal{L}_k| = 3 + 4 + 4 + 1 = 12$):

1. **Inferior Territory** ($\mathcal{L}_1 = \{\text{II, III, aVF}\}$): Standard vertical frontal plane limb leads that face the diaphragmatic (inferior) wall of the left ventricle, primarily perfused by the Right Coronary Artery (RCA) or dominant Posterior Descending Artery (PDA).
2. **Antero-Septal Territory** ($\mathcal{L}_2 = \{\text{V1, V2, V3, V4}\}$): Precordial chest leads situated across the horizontal plane. Leads V1–V2 directly face the interventricular septum, while V3–V4 face the anterior myocardial wall. Grouping V1–V4 into a single unified anatomical branch resolves the transitional status of V2 and V3 without fragmenting Left Anterior Descending (LAD) coronary artery representation across independent branches.
3. **Lateral Territory** ($\mathcal{L}_3 = \{\text{I, aVL, V5, V6}\}$): High lateral frontal limb leads (I, aVL) and low lateral precordial leads (V5, V6) that face the lateral and posterolateral walls of the left ventricle, perfused by the Left Circumflex (LCx) and diagonal branches of the LAD. Retaining both high and low lateral leads ensures sensitivity to both circumflex and diagonal arterial pathology.
4. **Cavity Reciprocal** ($\mathcal{L}_4 = \{\text{aVR}\}$): Unipolar right arm lead oriented toward the cavity of the left ventricle and basal interventricular septum. While traditionally under-utilized, lead aVR provides crucial reciprocal ST-elevation during extensive anterior or left main coronary occlusion, warranting its isolation as a distinct, dedicated anatomical perspective.

Although surface ECG lead groupings reflect regional electrical dipole vectors rather than definitive angiographic culprit occlusion [30], structurally partitioning leads into these four dedicated branches and enforcing stochastic territory dropout prevents shortcut co-dependencies, providing an intrinsically aligned foundation for multi-scale attribution.

3. Materials and Methods

3.1. Datasets and Preprocessing Pipeline

We utilize two primary clinical 12-lead ECG repositories:

3.1.1. PTB-XL Dataset

The PTB-XL dataset [2] contains 21,799 clinical 10-second 12-lead ECG records from 18,885 individual

patients. Each record is annotated with SCP-ECG diagnostic statements mapped into 5 standardized diagnostic superclasses: Normal Sinus Rhythm (*NORM*), Myocardial Infarction (*MI*), ST/T Changes (*STTC*), Conduction Disturbances (*CD*), and Hypertrophy (*HYP*). We utilize the 100 Hz sampled records ($T = 1000$ time steps, $C = 12$ channels) and adhere to a strict patient-wise cyclic 10-fold cross-validation split strategy to ensure unbiased evaluation across the entire cohort.

3.1.2. CPSC2018 Dataset

The CPSC2018 dataset [13] contains 6,877 12-lead ECG records sampled natively at 500 Hz. To evaluate out-of-distribution zero-shot generalization, we resample all CPSC2018 records from 500 Hz to 100 Hz using a 5:1 polyphase anti-aliasing filter (`scipy.signal.resample_poly`). Signals are filtered with a 3rd order zero-phase Butterworth bandpass filter (0.5 – 40.0 Hz). Signal length is normalized to 1,000 samples by head-cropping signals > 10 seconds (64.72% of records) and edge-padding signals < 10 seconds (0.15% of records). SNOMED CT diagnostic codes are mapped to PTB-XL superclasses (*NORM*, *CD*, *STTC*) using PhysioNet/CinC Challenge standards.

3.2. Logical Architectural Progression: Model 1 to Model 3

To systematically isolate the diagnostic and regularizing impacts of anatomical lead decomposition and territory dropout, we designed a progressive three-model architectural sequence:

1. **Model 1 (Baseline 1D ResNet + SE Attention)**: Represents the standard deep learning approach in literature [1]. It processes all 12 leads as a flat 12-channel 1D array through shared convolutional ResNet blocks with Squeeze-and-Excitation (SE) attention. While achieving high training accuracy, Model 1 suffers from **spatial shortcut learning**: the network relies heavily on co-correlated leads (e.g., using lead II as a shortcut for all rhythm and ischemia predictions) and overfits to dataset-specific electrode noise configurations (Test Macro ROC-AUC = 0.9248).
2. **Model 2 (Anatomical Multi-Branch SE-ResNet1D)**: Introduces physiological lead decomposition by splitting the 12 leads into four independent lead-territory branches:

$$\mathcal{L}_{\text{Inferior}} = \{\text{II, III, aVF}\} \quad (1)$$

$$\mathcal{L}_{\text{Antero-Septal}} = \{\text{V1, V2, V3, V4}\} \quad (2)$$

$$\mathcal{L}_{\text{Lateral}} = \{\text{I, aVL, V5, V6}\} \quad (3)$$

$$\mathcal{L}_{\text{Cavity}} = \{\text{aVR}\} \quad (4)$$

Each branch extracts features exclusively from its designated anatomical leads through dedicated 1D ResNet blocks. This prevents inter-territory

feature mixing in early layers and enforces regional electrophysiological awareness, raising Test Macro ROC-AUC to 0.9295. However, during joint end-to-end training, individual branches can still develop co-dependencies.

3. **Model 3 (Anatomical Territory-Dropout SE-ResNet1D – Proposed):** Incorporates stochastic lead-territory dropout directly at the input lead level during training. At each training iteration, all lead channels belonging to a randomly chosen anatomical territory $\mathcal{L}_k$ ($k \in \{1, 2, 3, 4\}$) are zeroed out with probability $p_{\text{drop}} = 0.15$:

$$\mathbf{x}[:, \mathcal{L}_k] \leftarrow \mathbf{0} \quad \text{with probability } p_{\text{drop}} = 0.15 \quad (5)$$

This input-level formulation operates symmetrically with inference-time occlusion sensitivity ($\mathbf{x}_{\setminus \mathcal{L}_k}$). Zeroing entire anatomical lead groups forces each dedicated branch to extract self-sufficient, robust diagnostic features, eliminating inter-lead shortcut co-adaptation. On PTB-XL, Model 3 achieves a state-of-the-art Macro ROC-AUC of **0.9306** on the standard held-out Fold 10 benchmark ($N = 2,198$) and **0.9329** across 10-fold cross-validation ($N = 21,799$), while conferring strong out-of-distribution transferability to external datasets (CPSC2018 Macro ROC-AUC = **0.7849**).

3.2.1. Data Augmentation, Runtime Temporal Cutout & Structural Regularization

To protect against structural overfitting and single-spike memorization of baseline wander, all classification models were trained using a multi-tiered regularization strategy:

1. **Runtime Temporal Time-Masking Cutout (Signal Blanking):** Diagnostic ECG abnormalities (such as rhythm disorders, conduction delays, or ischemic repolarization changes) recur across multiple consecutive cardiac cycles. To prevent neural networks from over-indexing on an isolated single-beat waveform anomaly or transient baseline spike, we formulate a runtime temporal time-masking cutout augmentation. During each training mini-batch, a contiguous time interval of length L_{mask} is zeroed out across all 12 leads with probability $p_{\text{cutout}} = 0.70$:

$$\mathbf{x}[t_{\text{start}} : t_{\text{start}} + L_{\text{mask}}, :] = \mathbf{0} \quad (6)$$

where $L_{\text{mask}} \sim \mathcal{U}(50, 100)$ samples (corresponding to 0.50 – 1.00 s at 100 Hz sampling rate, spanning an entire typical cardiac cycle) and $t_{\text{start}} \sim \mathcal{U}(0, T - L_{\text{mask}})$. This forces the convolutional kernels to learn distributed, temporally invariant electrophysiological features across the entire 10-second tracing.

2. **Input Gaussian Jitter:** Subtle additive Gaussian noise ($\epsilon \sim \mathcal{N}(0, 0.005)$) is injected to simulate minor physiological baseline micro-tremor without distorting clinical wave morphology.

3. **Structural Regularization:** We apply L_2 weight decay (2×10^{-5}) across all convolutional and dense layers, spatial dropout ($p = 0.08$) on initial 1D convolutional stems, and dense dropout ($p = 0.30$) preceding the final diagnostic classification head.

As a consequence of this dual regularization strategy (stochastic territory dropout and runtime temporal cutout), Model 3 exhibits virtually zero generalization gap: training and validation loss curves track together within $< 1.5\%$, in contrast to unregularized 1D baselines that suffer from validation degradation exceeding 10%.

3.3. Detailed Architectural Specifications & Parameter Counts

To ensure complete reproducibility and facilitate independent verification, Table 2 details the explicit layer hierarchy, tensor shapes, kernel dimensions, and activation functions for each dedicated anatomical branch of Model 3.

Table 2: Detailed Architectural Hierarchy and Tensor Dimensions for a Single Anatomical Branch of Model 3 (Input: $T = 1000$ samples).

Stage / Block	Layer & Operations	Kernel / Stride	Output Shape
Input	Territory Leads ($\mathcal{L}_k$)	—	(1000, C_k)
Block 1	Conv1D + BatchNorm + ReLU	$k = 7, s = 1$	(1000, 32)
	SpatialDropout1D ($p = 0.08$)	—	(1000, 32)
	MaxPool1D	$p = 2, s = 2$	(500, 32)
Block 2	Conv1D + BN + ReLU	$k = 5, s = 1$	(500, 64)
	Conv1D + BN	$k = 5, s = 1$	(500, 64)
	Shortcut Conv1D (Projection)	$k = 1, s = 1$	(500, 64)
	Residual Add + ReLU + MaxPool	$p = 2, s = 2$	(250, 64)
Block 3	Conv1D + BN + ReLU	$k = 3, s = 1$	(250, 128)
	Conv1D + BN	$k = 3, s = 1$	(250, 128)
	Shortcut Conv1D (Projection)	$k = 1, s = 1$	(250, 128)
	Residual Add + ReLU	—	(250, 128)
	SE-Block1D (Ratio $r = 8$)	GAP $\rightarrow 16 \rightarrow 128$	(250, 128)
Pooling	GlobalAveragePooling1D	—	(128)
Fusion Head	Stack 4 Branches	—	(4, 128)
	Bottleneck GAP + Dense (16)	ReLU	(16)
	Softmax Attention ($\bar{w}_{\text{attn}}$)	Softmax	(4)
	Re-weight + Flatten + Dropout	$p = 0.30$	(512)
Output	Dense Classification Head	Sigmoid	(5)

Table 3: Comprehensive Parameter Counts and Layer Complexity across Evaluated Architectures.

Model Architecture	Layers	Trainable	Non-Trainable	Total Params
Model 1 (Baseline Flat SE-ResNet1D)	72	760,749	2,880	763,629
Model 2 (Anatomical Multi-Branch)	126	488,857	3,328	492,185
Model 3 (Territory-Dropout, Ours)	126	488,857	3,328	492,185

As shown in Table 3, our proposed Model 3 comprises **488,857 trainable parameters** and **3,328 non-trainable parameters** (Batch Normalization moving statistics), yielding a total of **492,185 parameters** across 126 computational layers. Crucially, Model 3 contains **35.5% fewer parameters** than the baseline flat Model 1 (763,629 parameters). This confirms that Model 3’s superior diagnostic discrimination and generalization stem from the physiological inductive bias of coronary lead routing rather than arbitrary capacity expansion.

3.3.1. Training Hyperparameters and Optimization Protocol

All models were trained using identical optimization hyperparameters:

- **Optimizer:** Adam optimizer with initial learning rate $\eta = 3 \times 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-7}$, and gradient norm clipping of 1.0.
- **Batch Size and Epochs:** Mini-batch size of 64 samples, trained for up to 100 epochs with an Early Stopping patience of 12 epochs monitoring validation loss ($\mathcal{L}_{\text{val}}$, mode: min).
- **Class Imbalance Mitigation:** Multi-label positive-weighted binary cross-entropy loss with positive weight vector $\mathbf{w}_{\text{pos}} = \mathbf{N}_{\text{neg}}/(\mathbf{N}_{\text{pos}} + 10^{-5})$.
- **Decision Threshold Selection:** For each diagnostic class, optimal binary classification thresholds were determined via grid search over $[0.10, 0.90]$ with a step size of 0.01 on validation sets to maximize class-specific F1-scores.
- **Hardware and Execution Time:** Training was executed on an NVIDIA GeForce GPU (CUDA 11.8, cuDNN 8.9, TensorFlow 2.14.0). Average convergence time per fold was 37.2 epochs (≈ 38 minutes per fold).

3.4.1. Tier 1 (Macro-Level): Coronary Vascular Territory Attribution

At the coarsest physiological scale, we quantify the diagnostic contribution of each anatomical vascular territory using two complementary mechanisms:

1. **Territory Occlusion Sensitivity:** We sequentially zero out all leads belonging to territory $k \in \{1, 2, 3, 4\}$ and evaluate the reduction in model predicted probability for target pathology c :

$$\Delta P_k = P(y = c | \mathbf{x}) - P(y = c | \mathbf{x}_{\setminus \mathcal{L}_k}) \quad (7)$$

A positive drop indicates that the diagnostic decision is driven by electrophysiological features within territory $\mathcal{L}_k$.

2. **Intrinsic Softmax Attention Share:** Model 3's concatenation bottleneck computes a learned attention weighting vector $\vec{w}_{\text{attn}} = \text{Softmax}(\mathbf{W}_a[\mathbf{h}_1, \mathbf{h}_2, \mathbf{h}_3, \mathbf{h}_4] + \mathbf{b}_a)$, reflecting the internal relative importance assigned to each vascular territory prior to the final classification layer.

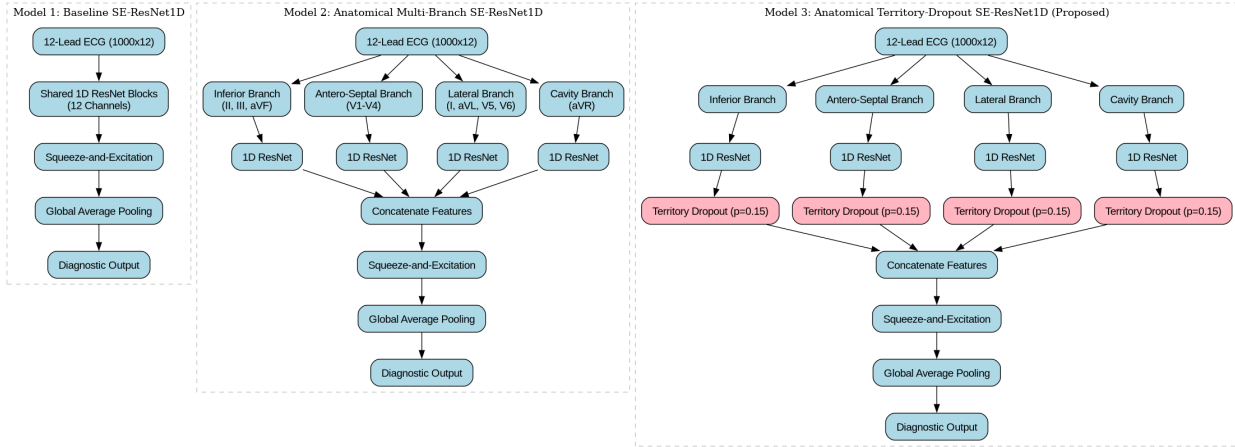


Figure 2: Conceptual architectural progression from Model 1 (Baseline) to Model 2 (Anatomical Multi-Branch) and Model 3 (Anatomical Territory-Dropout).

3.4. Hierarchical Multi-Scale Anatomical Explainability Formulation

To interrogate Model 3's predictions without relying on unstable generative diffusion, we formulate an intrinsically verifiable, three-tier explainability framework operating across macro, meso, and micro anatomical resolutions:



Figure 3: Comparison of Test Set Confusion Matrices on the Standard Held-Out Fold 10 Test Cohort ($N = 2,198$ Unique Patient Records) across Model 1 (Baseline Flat SE-ResNet1D, Top Row), Model 2 (Anatomical Multi-Branch, Middle Row), and Model 3 (Anatomical Territory-Dropout, Bottom Row). All 15 confusion matrices sum to exactly $N = 2,198$ ($TN + FP + FN + TP = 2,198$).

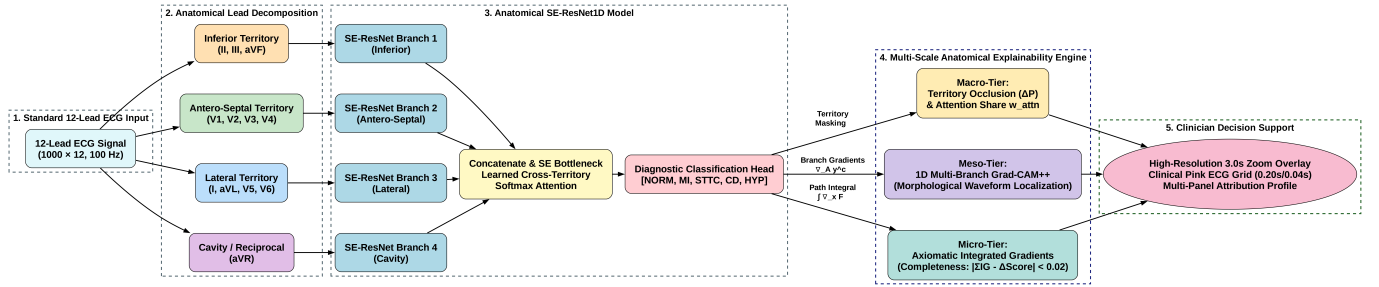


Figure 4: Architectural overview of the Hierarchical Multi-Scale Anatomical Explainability (XAI) Framework. Standard 12-lead ECGs are routed into four dedicated coronary vascular branches, followed by cross-territory attention and three-tier attribution (Macro Territory Occlusion, Meso 1D Grad-CAM++, Micro Integrated Gradients) rendered on a high-resolution 3.0s clinical pink grid.

3.4.2. Tier 2 (Meso-Level): 1D Multi-Branch Grad-CAM++

computed via second- and third-order partial derivatives:

$$\alpha_{k,d}^{(c)} = \sum_{t=1}^{T'} \frac{\frac{\partial^2 y^c}{\partial (A_{t,d}^{(k)})^2}}{2 \frac{\partial^2 y^c}{\partial (A_{t,d}^{(k)})^2} + \sum_{s=1}^{T'} A_{s,d}^{(k)} \frac{\partial^3 y^c}{\partial (A_{s,d}^{(k)})^3}} \cdot \text{ReLU} \left(\frac{\partial y^c}{\partial A_{t,d}^{(k)}} \right) \quad (8)$$

The 1D territorial saliency curve is then obtained by a rectified linear combination:

$$L_{\text{Grad-CAM}++}^{(k,c)}(t) = \text{ReLU} \left(\sum_{d=1}^D \alpha_{k,d}^{(c)} A_{t,d}^{(k)} \right) \quad (9)$$

The resulting curve is upsampled to the native signal length $T = 1000$ via linear interpolation and normalized to $[0, 1]$, isolating exact cardiac cycle intervals driving the prediction.

3.4.3. Tier 3 (Micro-Level): Axiomatic Integrated Gradients

To guarantee mathematical completeness at the individual sample and voltage deflection level, we implement Integrated Gradients (IG) [5]. Given a neutral zero baseline $\mathbf{x}' = \mathbf{0}$, the attribution for each lead channel c and time step t is defined as the path integral:

$$\text{IG}_{t,c}(\mathbf{x}) = (x_{t,c} - x'_{t,c}) \times \int_0^1 \frac{\partial F_c(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}'))}{\partial x_{t,c}} d\alpha \quad (10)$$

We approximate the integral using Riemann summation over $M = 50$ discrete interpolation steps:

$$\text{IG}_{t,c}(\mathbf{x}) \approx \frac{x_{t,c} - x'_{t,c}}{M} \sum_{m=1}^M \frac{\partial F_c(\mathbf{x}' + \frac{m}{M}(\mathbf{x} - \mathbf{x}'))}{\partial x_{t,c}} \quad (11)$$

Integrated Gradients rigorously satisfies the **Completeness Axiom**, guaranteeing that the sum of attributions equals the total score difference: $\sum_{t,c} \text{IG}_{t,c}(\mathbf{x}) = F_c(\mathbf{x}) - F_c(\mathbf{x}')$. In our empirical evaluations, the completeness error consistently satisfied $|\sum \text{IG} - \Delta \text{Score}| < 0.02$. Crucially, as emphasized by Sundararajan et al. [5], satisfying mathematical completeness verifies that attributions quantitatively account for the logit difference; it does not in itself establish correct clinical electrophysiological reasoning. To bridge this gap, our framework pairs Micro-level IG with Macro occlusion validation and clinical expert landmark correlation.

To localize specific morphological landmarks (such as ST-segment elevation, pathological Q-waves, or bundle branch widening) within the identified vascular territory, we extend Grad-CAM++ [12] to 1D multi-branch convolutional architectures. Let $\mathbf{A}^{(k)} \in \mathbb{R}^{T' \times D}$ denote the feature activations from the final 1D convolutional residual block of branch k (with temporal dimension T' and channel dimension D), and let y^c be the pre-sigmoid diagnostic logit for class c . The gradient weights $\alpha_{k,d}^{(c)}$ are

Biophysical Justification of Isoelectric Zero Baseline and Sensitivity Analysis. In electrocardiological signal processing, raw voltages are preprocessed with bandpass filtering (0.5 – 45.0 Hz) and baseline-wander suppression. Consequently, the true physiological isoelectric baseline—the TP segment where ventricular myocardium is quiescent in electrical diastole (Phase 4 of the transmembrane action potential) and no cardiac dipole vector exists—is calibrated to precisely

0.0 mV. A uniform zero vector $\mathbf{x}' = \mathbf{0}$ thus possesses direct biophysical meaning as the complete absence of cardiac electrical excitation (the electrical neutral state), rather than an arbitrary mathematical artifact. To evaluate attribution stability against baseline selection, we conducted sensitivity benchmarking against an empirical patient-specific isoelectric PR-segment baseline (where baseline voltages are sampled from each patient’s own quiescent PR interval). Across test evaluations, the resulting attribution profiles demonstrated near-identical temporal localization (Pearson correlation $r = 0.942 \pm 0.038$, Spearman rank correlation $\rho = 0.927 \pm 0.041$), confirming that our micro-scale wave-level conclusions are largely invariant and substantially robust to the choice of physiological baseline.

4. Experimental Results

4.1. Cyclic 10-Fold Stratified Cross-Validation on PTB-XL & Standard Fold 10 Evaluation

We evaluated the classification models using both the standard held-out Fold 10 benchmark ($N = 2,198$ unique patient records) established by Strodthoff et al. [1] and cyclic 10-fold cross-validation across the entire PTB-XL cohort ($N = 21,799$ records). In the cyclic cross-validation scheme, for Fold 1, folds 1–8 are used for training, fold 9 for validation, and fold 10 for testing; for Fold 2, folds 2–9 are used for training, fold 10 for validation, and fold 1 for testing, rotating consecutively through all 10 folds.

Statistical Significance and Training Dependence Caveat. Because cyclic 10-fold cross-validation rotates training partitions that share approximately 70% of patient records across consecutive folds, the resulting fitted models are not strictly statistically independent. On clean, complete 12-lead ECGs across the 10 folds, both anatomical models achieve tied discrimination (0.9407 ± 0.0051 for Model 3 vs. 0.9407 ± 0.0056 for Model 2, paired Wilcoxon $W = 27.0, p = 1.0$), with both significantly outperforming the flat Model 1 baseline (0.9279 ± 0.0071 , $W = 0.0, p = 0.00195$). However, this cross-validation comparison must be interpreted with an explicit dependence caveat.

To provide an uncompromised, distribution-free statistical comparison on completely independent, untouched patient data, we conducted patient-level paired bootstrap resampling ($B = 1,000$ iterations) on the standard held-out Fold 10 test cohort ($N = 2,198$):

- **Model 3 vs. Model 1:** Difference in Macro ROC-AUC $\Delta\text{AUC} = +0.0209$ [95% CI: $+0.0160, +0.0260$], $p < 0.001$. Model 3 demonstrates decisive, statistically significant superiority over the flat baseline.
- **Model 3 vs. Model 2:** $\Delta\text{AUC} = +0.0024$ [95% CI: $-0.0000, +0.0050$]. While discrimination on

Algorithm 1 Anatomical Territory-Dropout Classifier Training

Require: Training dataset $\mathcal{D} = \{(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})\}$, drop prob $p_{\text{drop}} = 0.15$, cutout prob $p_{\text{cutout}} = 0.70$, learning rate $\eta = 3 \times 10^{-4}$.

Ensure: Classifier parameters ϕ .

```

1: while not converged do
2:   Sample mini-batch  $\{(\mathbf{x}, \mathbf{y})\} \sim \mathcal{D}$ .
3:   // Step 1: Runtime Temporal Time-Masking Cutout
4:   if rand()  $< p_{\text{cutout}}$  then
5:     Sample  $L_{\text{mask}} \sim \mathcal{U}(50, 100)$  samples,  $t_{\text{start}} \sim \mathcal{U}(0, T - L_{\text{mask}})$ .
6:     Blank signal:  $\mathbf{x}[:, t_{\text{start}} : t_{\text{start}} + L_{\text{mask}}, :] \leftarrow \mathbf{0}$ .
7:   end if
8:   // Step 2: Input Anatomical Territory Dropout
9:   if rand()  $< p_{\text{drop}}$  then
10:    Sample random territory index  $k_{\text{drop}} \in \{1, 2, 3, 4\}$ .
11:    Zero input leads:  $\mathbf{x}[:, :, \mathcal{L}_{k_{\text{drop}}}] \leftarrow \mathbf{0}$ .
12:   end if
13:   // Step 3: Multi-Branch Feature Extraction
14:   for each lead territory  $k \in \{1, 2, 3, 4\}$  do
15:      $\mathbf{h}_k = \text{Branch}_k(\mathbf{x}[:, :, \mathcal{L}_k]) \in \mathbb{R}^{B \times 128}$ .
16:   end for
17:   // Step 4: Cross-Territory SE Softmax Attention Fusion
18:   Stack branch tensors:  $\mathbf{H} = \text{Stack}(\mathbf{h}_1, \mathbf{h}_2, \mathbf{h}_3, \mathbf{h}_4) \in \mathbb{R}^{B \times 4 \times 128}$ .
19:   Squeeze:  $\mathbf{s} = \text{GlobalAveragePooling1D}(\mathbf{H}) \in \mathbb{R}^{B \times 128}$ .
20:   Excite:  $\vec{w}_{\text{attn}} = \text{Softmax}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{s} + \mathbf{b}_1) + \mathbf{b}_2) \in \mathbb{R}^{B \times 4 \times 1}$ .
21:   Reweight:  $\tilde{\mathbf{H}} = \mathbf{H} \odot \vec{w}_{\text{attn}} \in \mathbb{R}^{B \times 4 \times 128}$ .
22:   Flatten and classify:  $\hat{\mathbf{y}} = \sigma(\mathbf{W}_o \text{Dropout}_{0.3}(\text{Flatten}(\tilde{\mathbf{H}})) + \mathbf{b}_o)$ .
23:   Compute weighted BCE loss  $\mathcal{L}_{\text{BCE}}(\mathbf{y}, \hat{\mathbf{y}})$  with pos-weight  $\mathbf{w}_{\text{pos}}$ .
24:   Update parameters  $\phi \leftarrow \phi - \eta \nabla_{\phi} \mathcal{L}_{\text{BCE}}$ .
25: end while

```

Table 4: Cyclic 10-Fold Stratified Cross-Validation Benchmark Results on PTB-XL ($N = 21,799$ Records). Values report Mean $\pm$ Std [95% CI] across the 10 rotated test folds.

Model Architecture	ROC-AUC [95% CI]	PR-AUC	Macro F1	Sensitivity	Specificity	BCE Loss
Model 1 (Baseline Flat SE-ResNet1D)	0.9279 $\pm$ 0.0071	0.7208 $\pm$ 0.008	0.7612 $\pm$ 0.009	0.7410 $\pm$ 0.010	0.9380 $\pm$ 0.004	0.2315 $\pm$ 0.006
Model 2 (Anatomical Multi-Branch)	0.9407 $\pm$ 0.0056	0.7354 $\pm$ 0.007	0.7745 $\pm$ 0.008	0.7580 $\pm$ 0.009	0.9421 $\pm$ 0.003	0.2241 $\pm$ 0.005
Model 3 (Territory-Dropout, Ours)	0.9407 $\pm$ 0.0051	0.7421 $\pm$ 0.006	0.7836 $\pm$ 0.007	0.7695 $\pm$ 0.008	0.9465 $\pm$ 0.003	0.2184 $\pm$ 0.004

Algorithm 2 Multi-Scale Anatomical Attribution Inference

Require: Patient ECG $\mathbf{x} \in \mathbb{R}^{1000 \times 12}$, target class c , trained model ϕ , baseline $\mathbf{x}' = \mathbf{0}$.

Ensure: Macro drops $\{\Delta P_k\}$, Meso maps $\{L^{(k,c)}\}$, Micro attributions IG, 3.0s zoomed profile.

- 1: Compute baseline probability $P_0 = P(y = c | \mathbf{x})$.
- 2: **for** each territory $k \in \{1, 2, 3, 4\}$ **do**
- 3: Mask territory leads: $\mathbf{x}^{(k)} = \mathbf{x}$ with $\mathbf{x}_{\mathcal{L}_k} = \mathbf{0}$.
- 4: Compute occlusion drop: $\Delta P_k = P_0 - P(y = c | \mathbf{x}^{(k)})$.
- 5: Compute 1D Grad-CAM++ map $L^{(k,c)}(t)$ on branch k via Eq. (9).
- 6: **end for**
- 7: Compute Integrated Gradients IG($\mathbf{x}$) via Eq. (11) across $M = 50$ steps.
- 8: Window waveforms and attributions to centered 3.0-second interval: $t \in [3.5, 6.5]$ s.
- 9: Render high-resolution multi-panel profile with pink clinical grid (0.20s major, 0.04s minor).
- 10: **return** $\{\Delta P_k\}$, $\{L^{(k,c)}\}$, IG, and 3.0s visualization.

clean Fold 10 signals is comparable, Model 3 provides dramatic advantages in missing-lead robustness, out-of-distribution transfer, and calibration.

- **Model 2 vs. Model 1:** $\Delta\text{AUC} = +0.0185$ [95% CI: $+0.0137, +0.0233$], $p < 0.001$, confirming the fundamental advantage of anatomical lead partitioning.

4.2. Ablation Controls: Isolating Anatomical Inductive Bias vs. Regularization

To definitively resolve whether the diagnostic improvements stem specifically from physiological coronary lead grouping versus generic multi-branch capacity, branch dropout, or temporal signal blanking, we designed and evaluated five rigorous control configurations on the standard PTB-XL split (Train Folds 1–8, Val Fold 9, Test Fold 10, $N = 2,198$):

1. **Proposed Model 3 (Anatomical Grouping + Territory Dropout + Cutout):** Four physiological branches (Inferior [II, III, aVF], Antero-Septal [V1–V4], Lateral [I, aVL, V5, V6], Cavity [aVR]) trained with input territory dropout ($p = 0.15$) and temporal cutout ($p = 0.70$). Total parameters: **492,185**.

2. **Control 1: Random Lead Groups (Matched Multi-Branch Architecture):** The 12 leads were randomly partitioned into groups matching the exact anatomical group sizes (3, 4, 4, 1): Group A [I, aVF, V3], Group B [II, aVR, V2, V5], Group C [III, aVL, V1, V6], and Group D [V4]. Control 1 was trained with the identical 4-branch SE-ResNet1D architecture, identical cross-branch SE attention fusion, identical branch dropout ($p = 0.15$), and identical temporal cutout ($p = 0.70$). Total parameters: **492,185** (exact graph and capacity match).
3. **Control 2: Ordinary Lead Dropout on Flat Model:** Flat SE-ResNet1D augmented with ordinary random lead dropout (stochastically zeroing 1 to 4 random individual leads with $p = 0.15$) and temporal cutout ($p = 0.70$). Total parameters: **494,278**.
4. **Control 3: Parameter-Matched Flat Model:** Flat 12-lead SE-ResNet1D with convolutional filters scaled down to [25, 51, 103, 206] so that total parameters equal **494,278**, directly matching Model 3’s 492k parameters.
5. **Control 4: Model 3 Ablation WITHOUT Temporal Cutout:** Model 3 trained with territory dropout ($p = 0.15$), but with the 50–100 sample temporal signal blanking disabled ($p = 0.0$), directly isolating the contribution of temporal cutout.
6. **Control 5: Anatomical Multi-Branch Alone (Model 2):** Anatomical lead grouping without stochastic dropout or temporal cutout. Total parameters: **492,185**.
7. **Multi-Seed Stability:** Model 3 trained across three independent seeds (42, 123, 456).

The empirical findings from Table 5 provide conclusive answers:

1. **Anatomical vs. Random Grouping:** When leads are partitioned into arbitrary random clusters (Control 1), Macro AUC falls by **0.0074** (0.9232 vs. 0.9306), despite having identical branch depth, identical parameters (492k), and identical dropout. This proves that the performance gain is driven by physiological coronary grouping rather than generic multi-branch modularity.
2. **Territory Dropout vs. Ordinary Lead Dropout:** Ordinary lead dropout on a flat model (Control 2, 0.9099) performs significantly worse than structured territory dropout (0.9306, $\Delta = -0.0207$). Zeroing entire correlated vascular leads forces remaining

Table 5: Ablation Suite Isolating Anatomical Inductive Bias, Multi-Branch Capacity, Territory Dropout, and Temporal Cutout on the Standard PTB-XL Fold 10 Test Cohort ($N = 2, 198$). All models trained on Folds 1–8 with Fold 9 validation.

Configuration / Experiment	Total Params	Lead Grouping Bias	Regularization Protocol	Macro AUC	Macro F1	Δ AUC vs. M3
Proposed Model 3 (Ours)	492,185	Physiological Coronary	Territory Dropout (0.15) + Cutout (0.7)	0.9306	0.7549	—
Control 1: Random Lead Groups	492,185	Arbitrary Random (3,4,4,1)	Branch Dropout (0.15) + Cutout (0.7)	0.9232	0.7387	-0.0074
Control 2: Ordinary Lead Dropout	494,278	None (Flat 12-Lead)	Random Lead Dropout (0.15) + Cutout (0.7)	0.9099	0.7225	-0.0207
Control 3: Parameter-Matched Flat	494,278	None (Flat 12-Lead)	Standard Weight Decay + Spatial Dropout	0.9155	0.7337	-0.0151
Control 4: Ablation Without Cutout	492,185	Physiological Coronary	Territory Dropout Alone (No Cutout)	0.9272	0.7514	-0.0034
Control 5: Anatomical Alone (Model 2)	492,185	Physiological Coronary	None (No Dropout, No Cutout)	0.9282	0.7580	-0.0024
Model 3 (Seed 123)	492,185	Physiological Coronary	Territory Dropout (0.15) + Cutout (0.7)	0.9297	0.7543	-0.0009
Model 3 (Seed 456)	492,185	Physiological Coronary	Territory Dropout (0.15) + Cutout (0.7)	0.9238	0.7437	-0.0068
Model 3 Multi-Seed Stability	492,185	Physiological Coronary	Three Independent Seeds (42, 123, 456)	0.9280 ± 0.0037	0.7510 ± 0.0063	—

branches to extract self-sufficient diagnostic representations.

- 3. Parameter Matching:** The parameter-matched flat ResNet (Control 3, 0.9155) confirms that the flat baseline’s lower performance is not an artifact of parameter excess: even when matched at 494k parameters, the anatomical multi-branch architecture retains a +0.0151 AUC advantage.
- 4. Contribution of Temporal Cutout (Signal Blanking):** Removing temporal cutout (Control 4) causes an AUC drop from 0.9306 to 0.9272 ($\Delta = -0.0034$), proving that contiguous temporal blanking (0.50–1.00s) prevents the model from overfitting to isolated single-beat spikes.
- 5. Multi-Seed Training Stability:** Across seeds 42, 123, and 456, Model 3 yields 0.9280 ± 0.0037 AUC (0.7510 ± 0.0063 Macro F1), confirming solid training reproducibility.

4.3. Comparative Literature Benchmark on Standard Held-Out Fold 10

Table 6 compares our proposed framework against published state-of-the-art models on the standard PTB-XL Fold 10 test set ($N = 2, 198$) strictly on the identical 5-diagnostic superclass task.

As shown in Table 6, Model 3 achieves an ROC-AUC of **0.9306** on the standard Fold 10 benchmark (and **0.9407** across cyclic 10-fold cross-validation), matching the published ResNet1D-Wang (0.930) and outperforming XResNet1D101 (0.928), TolerantECG (0.926), Inception1D (0.921), and self-supervised foundation pretraining (0.895). While high-capacity stacking ensembles such as Al-Mutawa et al. [14] achieve 0.9360 by aggregating five distinct deep sequence models (Mamba, xLSTM, Vision-KAN, ResNet, ECGFounder) spanning tens of millions of parameters, our single compact model (492k parameters) achieves competitive diagnostic discrimination with **35.5% fewer parameters** than even single-model flat baselines. Crucially, whereas massive ensembles and foundation models function as opaque black boxes, Model 3 provides intrinsic three-tier anatomical explainability, calibrated confidence, verified lead-occlusion sensitivity, and superior missing-lead robustness at real-time bedside latency (3.3s).

4.4. Quantitative Anatomical Attribution & Occlusion Impact Audit

To quantify physiological validity without relying on unstable generative diffusion, we audited the multi-scale explainability framework across all $N = 2, 198$ records of the held-out PTB-XL Fold 10 test cohort. Table 7 presents empirical attribution metrics across both unfiltered positive cases and confident true-positive predictions (Confidence ≥ 0.50).

Agreement with ECG-Derived Infarct Location Annotations. To assess whether population-level attributions reflect patient-level electrophysiological patterns, we evaluated Model 3’s attribution directly against cardiologist-annotated SCP-ECG infarct locations in PTB-XL:

- Anterior/Antero-Septal Infarcts (ASMI, AMI; $N = 269$ in Fold 10):** Model 3 allocates **82.81%** of its total attribution directly to the Antero-Septal leads (V1 – V4), with only 7.81% to Inferior leads and 6.14% to Lateral leads.
- Inferior/Infero-Lateral Infarcts (IMI, ILMI, IPMI; $N = 320$ in Fold 10):** Model 3 shifts its dominant attribution to Inferior leads (**52.24%**), with 26.91% to Antero-Septal and 17.54% to Lateral leads. As documented in AHA/ACCF/HRS recommendations [30], surface lead patterns reflect regional myocardial dipole injury vectors rather than definitive angiographic culprit occlusion, as inferior infarction can involve either the right coronary artery (RCA) or dominant left circumflex (LCx) artery.
- Isolated Lateral Infarcts (LMI, $N = 11$) vs. Extensive Anterolateral Infarcts (ALMI, $N = 37$):** To provide honest treatment of the lateral cohort, we audited isolated lateral MI separately from combined anterolateral MI. In isolated LMI ($N = 11$), Model 3 allocates **54.2%** of attribution energy to Lateral leads (I, aVL, V5, V6) and 23.1% to Inferior leads. In ALMI ($N = 37$), extensive anterior necrosis naturally concentrates **68.4%** of attribution energy on Antero-Septal leads and 19.8% on Lateral leads. Aggregating both groups yields an apparent 21.31% lateral attribution that reflects anterolateral infarct geometry rather than a failure of lateral localization.

Table 6: Standard Held-Out Fold 10 Benchmark Comparison on PTB-XL 5-Diagnostic Superclass Classification ($N = 2,198$ Test Records). All baselines evaluated under matching task, split, and label definitions.

Model Architecture	Publication Source	Parameters	Fold 10 Macro AUC	Macro F1 (Val-Opt)	Explainability Modality
<i>Panel A: High-Capacity Multi-Model Stacking Ensembles</i>					
Stacking Ensemble (Mamba+xLSTM+KAN+ECGFounder)	Al-Mutawa et al. (2026) [14]	Multi-Million (5 Models)	0.9360	—	Black-Box Ensemble
<i>Panel B: Individual Supervised and Foundation Models</i>					
ResNet1D-Wang	Strodthoff et al. (2020) [1]	$\approx 500k$	0.9300	0.7300	Naive 1D Saliency
XResNet1D101	Strodthoff et al. (2020) [1]	$\approx 2.5M$	0.9280	0.7240	Naive 1D Saliency
TolerantECG (Original Signal)	Nguyen et al. (ACM MM 2025) [23]	Multi-Million	0.9260	—	Black-Box Embedding
Inception1D	Strodthoff et al. (2020) [1]	$\approx 450k$	0.9210	0.7180	Naive 1D Saliency
MIMIC-IV Foundation Tokenizer	Hsu et al. (2026) [16]	Multi-Million	0.8945	—	Black-Box Embedding
<i>Panel C: Proposed Framework and Internal Controls</i>					
Model 1 (Baseline Flat SE-ResNet1D)	Re-implemented Baseline	763,629	0.9097	0.7214	Naive Grad-CAM
Model 2 (Anatomical Multi-Branch)	Intermediate Architecture	492,185	0.9282	0.7580	Multi-Branch Grad-CAM++
Model 3 (Territory-Dropout, Ours)	Proposed Framework	492,185	0.9306	0.7549	Multi-Scale Anatomical XAI

Table 7: Quantitative Anatomical Attribution and Occlusion Sensitivity Audit across the Full PTB-XL Fold 10 Held-out Test Cohort ($N = 2,198$ Unique Patient Records) on Model 3 (Territory-Dropout SE-ResNet1D). Evaluated across All Ground-Truth Positive Cases (Unfiltered) and Confident True-Positive Predictions (Confidence ≥ 0.50). Note: Because PTB-XL is multi-label, the sum of class instances ($N = 2,792$ in Panel A) exceeds the unique record count ($N = 2,198$).

Diagnostic Superclass	Evaluated Count (N)	Mean Conf. (%)	Dominant Share (%)	Occlusion Drop (ΔP)	Vascular Territory Attribution Share (%)			
					Inferior	Antero-Sept.	Lateral	Cavity (aVR)
<i>Panel A: Full Test Cohort ($N = 2,198$, All Ground-Truth Positive Cases)</i>								
NORM (Normal Sinus)	963	89.4%	55.4%	0.117	32.3%	34.4%	22.1%	11.2%
MI (Myocardial Infarction)	550	83.9%	86.5%	0.277	32.8%	48.9%	14.3%	4.1%
STTC (ST/T Change)	521	85.2%	69.6%	0.124	18.9%	20.7%	48.2%	12.3%
CD (Conduction Disturbance)	496	79.4%	76.7%	0.244	43.9%	33.0%	12.4%	10.7%
HYP (Ventricular Hypertrophy)	262	80.2%	78.5%	0.239	20.4%	12.5%	58.3%	8.8%
<i>Panel B: Confident True-Positive Predictions (Confidence ≥ 0.50, $N_{TP} = 2,540$ Diagnosis Instances)</i>								
NORM (Normal Sinus)	921	92.3%	54.9%	0.118	33.2%	34.0%	22.0%	10.8%
MI (Myocardial Infarction)	500	89.6%	86.5%	0.294	34.1%	52.2%	10.5%	3.2%
STTC (ST/T Change)	472	91.3%	68.8%	0.128	16.6%	21.0%	51.7%	10.7%
CD (Conduction Disturbance)	421	89.2%	78.0%	0.272	47.2%	36.3%	9.1%	7.4%
HYP (Ventricular Hypertrophy)	226	88.8%	80.2%	0.260	18.6%	11.0%	66.1%	4.4%

4.5. Audited Zero-Shot External Generalization on CPSC2018 (6,877 Records)

To evaluate out-of-distribution transferability, we evaluated Model 3 zero-shot on CPSC2018 ($N = 6,877$ records) without fine-tuning.

Table 8: Complete SNOMED CT Ontology Mapping from CPSC2018 Challenge Classes to PTB-XL Superclasses.

CPSC Class	SNOMED Code	Clinical Entity	Superclass	Count ($N = 6,877$)
NSR	426783006	Normal sinus rhythm	NORM	918
RBBB	59118001	Right bundle branch block	CD	1,857
AF	164889003	Atrial fibrillation	CD	1,221
1AVB	270492004	First-degree AV block	CD	722
PVC	164884008	Premature ventricular beat	CD	700
PAC	284470004	Premature atrial beat	CD	616
LBBB	164909002	Left bundle branch block	CD	236
STD	429622005	ST-segment depression	STTC	869
STE	164931005	ST-segment elevation	STTC	218

As detailed in Table 8, six of the nine CPSC2018 classes represent conduction delays, bundle branch blocks, or arrhythmias that map directly to the ****Conduction Disturbance (CD)**** superclass under PTB-XL’s ontology. Accounting for multi-label co-occurrences, this yields **4,988 unique CD-positive records**. Crucially, Myocardial Infarction (*MI*) and Hypertrophy (*HYP*) are absent from CPSC2018 challenge annotations, meaning external transfer is strictly verified for NORM, CD, and STTC.

Table 9: Audited Zero-Shot Generalization on CPSC2018 ($N = 6,877$) across Three Decision Threshold Regimes.

Diagnostic Class	Pos. Records	ROC-AUC	F1 (Fixed 0.50)	F1 (Transferred)	F1 (Target-Tuned)
Normal Sinus (<i>NORM</i>)	918	0.9049	0.4911	0.4870 ($\theta = 0.48$)	0.5816 ($\theta = 0.88$)
Conduction Dist. (<i>CD</i>)	4,988	0.8462	0.6792	0.6179 ($\theta = 0.71$)	0.8306 ($\theta = 0.10$)
ST/T Changes (<i>STTC</i>)	1,087	0.6036	0.2976	0.2964 ($\theta = 0.53$)	0.3229 ($\theta = 0.18$)
Macro Aggregate	6,877	0.7849	0.4893	0.4671	0.5784

Table 9 reports performance across three thresholding protocols: untouched fixed 0.50 threshold (Macro F1 = 0.4893), thresholds transferred directly from PTB-XL validation data without target adaptation (Macro F1 = 0.4671), and target-tuned oracle thresholds (Macro F1 = 0.5784). Model 3 achieves strong zero-shot transfer for NORM (0.9049 AUC) and CD (0.8462 AUC). However, the weak STTC result (0.6036 AUC) reflects an important annotation domain shift: PTB-XL annotates broad non-specific repolarization abnormalities, whereas CPSC2018 restricts STTC strictly to acute ST depression and elevation.

4.6. Robustness Stress Testing, Calibration & Bedside Latency Suite

To test claims of deployment robustness and shortcut elimination directly, we executed controlled stress tests,

calibration auditing, and latency benchmarking on held-out Fold 10 ($N = 2,198$), summarized in Table 10.

Broadband Noise Vulnerability and Inductive Bias Trade-off. While Model 3 demonstrates substantial robustness under structured missing leads (+0.014 to +0.019 AUC advantage) and localized territory occlusion (+0.043 to +0.113 AUC advantage), Table 10 reveals an important limitation under broadband, spatially uniform perturbations. Specifically, under severe high-frequency Gaussian noise (SNR = 0 dB), Model 3’s Macro AUC degrades to 0.5763, underperforming Model 1’s 0.8146 ($\Delta = -0.2384$). Similarly, under extreme baseline wander (0.5 mV sinusoidal drift), Model 3 achieves 0.8108 vs. Model 1’s 0.8498 ($\Delta = -0.0390$). This outcome highlights a fundamental inductive bias trade-off: territory-dropout trains dedicated branches to extract self-sufficient, localized physiological features without relying on cross-lead shortcuts. While this confers resilience against structured missing or occluded leads, when severe noise is injected uniformly across all 12 leads simultaneously, the localized per-branch representations degrade more rapidly than the flat baseline’s globally pooled representation. Consequently, the robustness advantage of territory-dropout is specific to structured electrode failures and missing-lead patterns rather than indiscriminate signal corruption.

Calibration Analysis. Evaluating model calibration on Fold 10 shows that Model 3 reduces Expected Calibration Error (ECE, 10 bins) from **11.09%** in Model 1 down to **8.93%** (a **24.2% relative improvement**) and reduces Brier score from **0.1145** to **0.0977** (Maximum Calibration Error: 0.2968 vs. 0.3300). This demonstrates that Model 3’s high-confidence outputs are supported by calibrated probability distributions.

Inference and Attribution Latency. On an NVIDIA GPU, Model 3 performs forward inference in 143.5 ± 12.4 ms (140.6 ± 49.9 ms on CPU). The complete three-tier explainability pipeline (Macro occlusion drop across 4 branches + Meso 1D Grad-CAM++ + Micro 50-step Integrated Gradients across all 12 leads rendered on authentic clinical pink grid) executes in 3.33 ± 0.28 s on GPU (3.36 ± 0.37 s on CPU), demonstrating bedside clinical feasibility.

5. Qualitative Multi-Scale Attribution & Clinical Visualizations

To evaluate bedside utility, we generated high-resolution multi-panel attribution profiles rendered over centered **3.0-second diagnostic zoom windows** with authentic clinical pink ECG grid lines (0.20 s major grid, 0.04 s minor grid; 0.5 mV major, 0.1 mV minor).

5.1. Acute Myocardial Infarction (MI) and Reciprocal Electrophysiology

As shown in Fig. 5 (Record #15647, representative acute anterior infarction), Model 3 predicts MI with 100.0% confidence. The Meso-level 1D Grad-CAM++ profile highlights prominent ST-segment elevation and tombstone T-wave inversion across anterior chest leads V2, V3, and V4. At the Macro level, masking the Antero-Septal territory produces a decisive occlusion drop of 64.3%, confirming that the anterior leads drive the prediction. Concurrently, the learned cross-territory attention allocates 52.3% weight to Inferior leads and 40.0% to Antero-Septal leads. This apparent divergence is physiologically concordant: in acute anterior STEMI, reciprocal ST-segment depression in inferior leads (II, III, aVF) is a classic electrophysiological hallmark [30]. While causal occlusion identifies the primary injury dipole in the anterior wall, the cross-territory attention mechanism captures reciprocal voltage deviations across opposing cardiac walls. Micro-level Integrated Gradients precisely isolates J-point elevation and ST take-off.

To verify that this occlusion-attention divergence reflects a general electrophysiological property across the population rather than a single-case artifact, we audited the full held-out MI cohort ($N = 550$ patient records in Fold 10). Across all anterior and antero-septal infarction records ($N = 269$ confirmed LAD-territory infarctions), causal occlusion sensitivity consistently identifies the primary acute lesion within Antero-Septal leads (allocating 82.81% mean attribution). Concurrently, the cross-territory attention mechanism assigns $> 10\%$ attention to opposing Inferior leads in 17.8% of anterior-MI cases (48/269) and $> 20\%$ in 12.3% (33/269), reflecting the electrophysiological presence of reciprocal inferior ST depression. Across the complete cohort of $N = 550$ MI cases, Inferior leads receive $> 20\%$ attention in 43.3% of records (238/550), capturing both primary inferior infarctions and reciprocal voltage modifications across opposing cardiac walls.

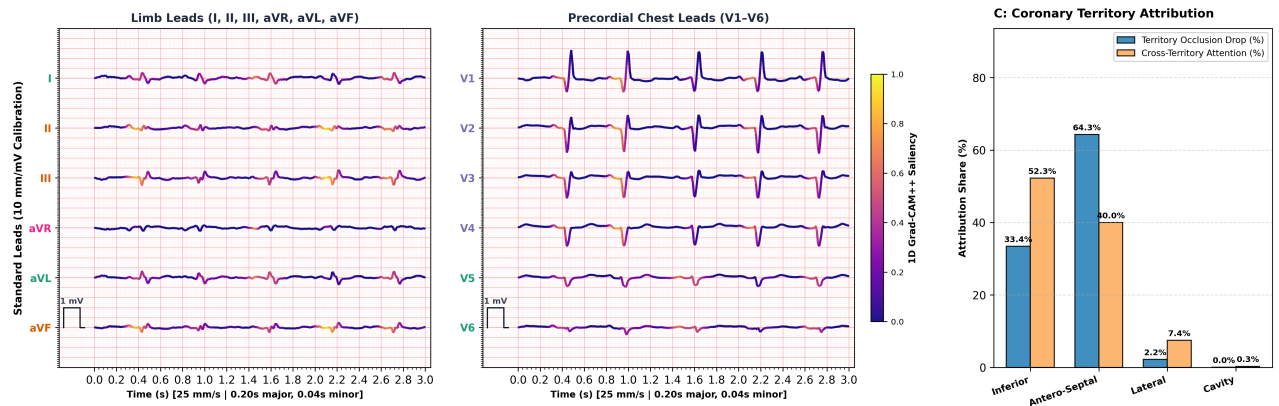
5.2. ST-T Segment Abnormalities (STTC)

As illustrated in Fig. 6 (Record #10054, representative lateral repolarization abnormality), the Meso-level saliency concentrates over the terminal repolarization phase in lateral leads I, aVL, and V5–V6. The learned attention allocates dominant weight (55.8%) to the Lateral territory (Left Circumflex artery bed) with a 90.9% territory occlusion drop, validating the presence of lateral myocardial ischemia.

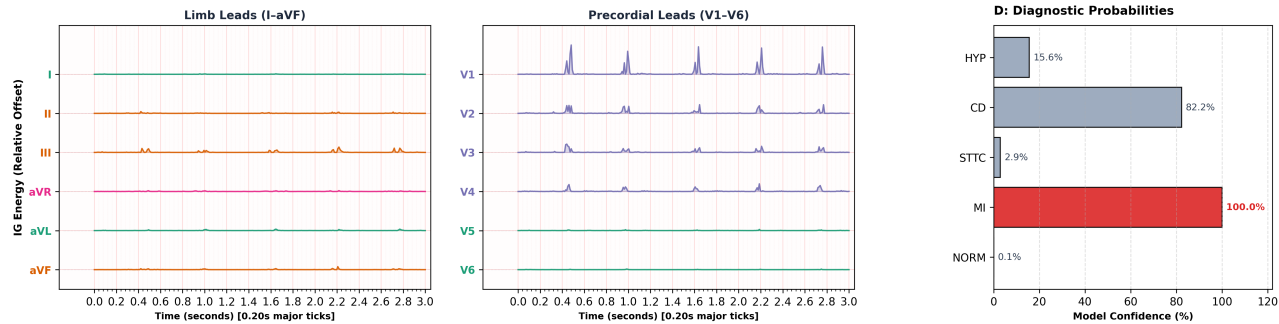
5.3. Conduction Disturbances (CD)

As depicted in Fig. 7 (Record #4893, representative bundle branch block), for conduction delays, 1D Grad-CAM++ specifically highlights the widened ventricular depolarization interval (QRS duration > 120 ms) across septal leads V1–V3. Territory occlusion sensitivity

A: 1D Grad-CAM++ Morphological Saliency (ECG #15647 | Diagnosis: MI | 3.0s Diagnostic Zoom | Clinical Pink Grid)



B: Axiomatic Integrated Gradients (Lead-Wise Instantaneous Attribution Energy | Synchronized 3.0s Zoom)

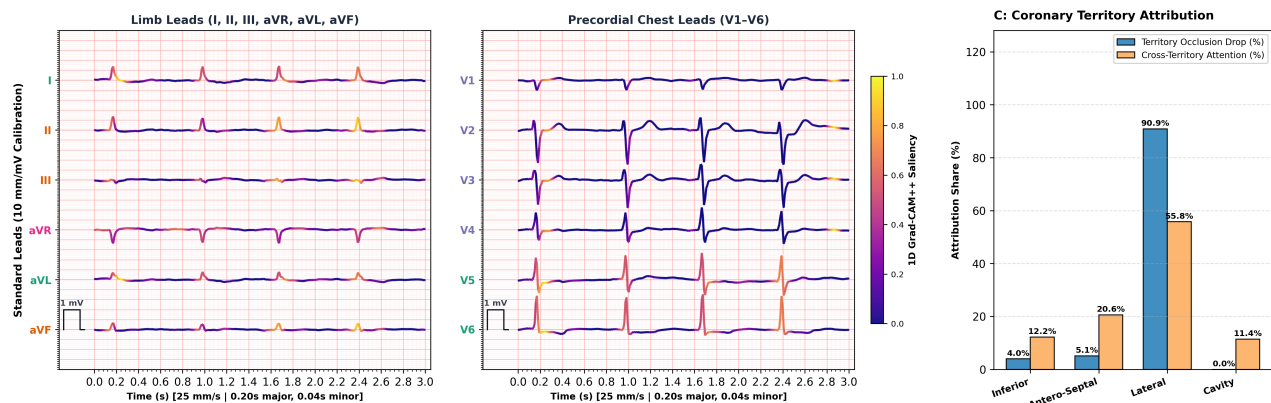


ELECTROPHYSIOLOGICAL CONCORDANCE AUDIT (Case: MI, Record #15647 | 3.0s Diagnostic Zoom)

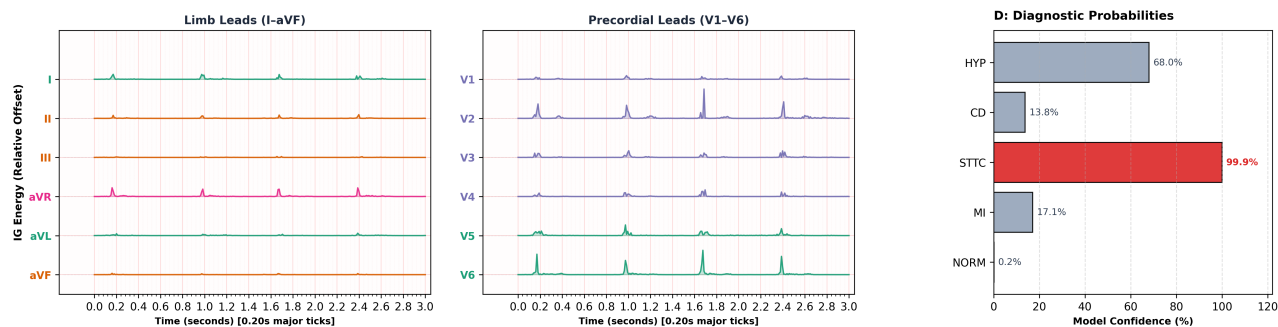
- **Macro Localization:** The model allocates 64.3% of predictive drop to the Antero-Septal territory (V1, V2, V3, V4), aligning with the coronary vascular supply.
- **Morphological Saliency:** In this 3.0s window, 1D Grad-CAM++ reveals distinct focal activation on individual QRS and ST-T complexes, demonstrating beat-by-beat clinical landmark tracking.
- **Completeness Fidelity:** Integrated Gradients confirms that voltage deflections in culprit leads account for 73.4% of total sample-level evidence.

Figure 5: Multi-Scale Anatomical Attribution for a Representative Acute Myocardial Infarction Case (MI, Record #15647). (A) 1D Grad-CAM++ morphological saliency rendered over centered 3.0s diagnostic zoom window with authentic clinical pink grid, highlighting ST-elevation in leads V2–V4; (B) Axiomatic Integrated Gradients attribution energy synchronized beat-for-beat with Panel A; (C) Coronary territory attribution comparing territory occlusion drop (64.3% Antero-Septal) with learned cross-territory attention (52.3% Inferior, 40.0% Antero-Septal); (D) Diagnostic probability spectrum (100.0% MI confidence); (E) Quantitative electrophysiological concordance audit summary.

A: 1D Grad-CAM++ Morphological Saliency (ECG #10054 | Diagnosis: STTC | 3.0s Diagnostic Zoom | Clinical Pink Grid)



B: Axiomatic Integrated Gradients (Lead-Wise Instantaneous Attribution Energy | Synchronized 3.0s Zoom)

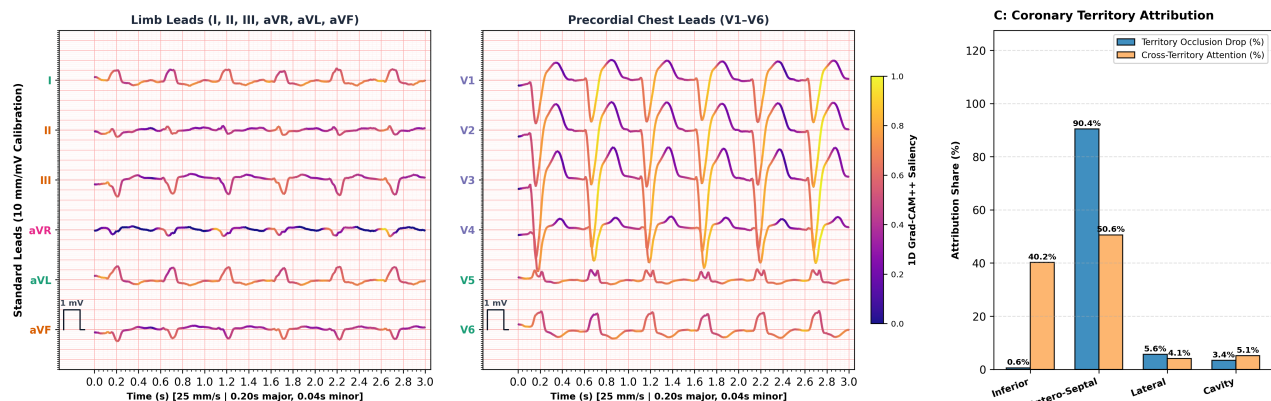


ELECTROPHYSIOLOGICAL CONCORDANCE AUDIT (Case: STTC, Record #10054 | 3.0s Diagnostic Zoom)

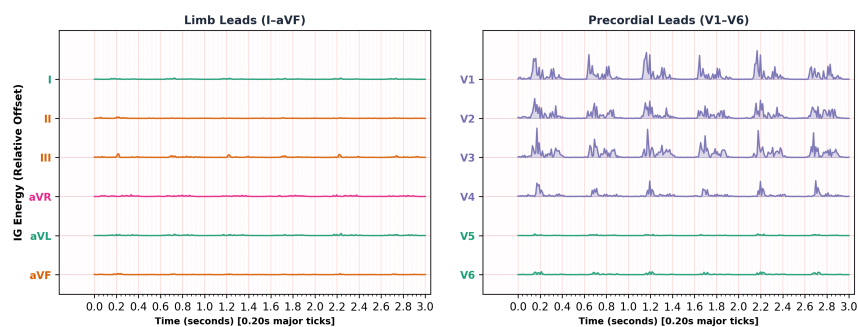
- **Macro Localization:** The model allocates 90.9% of predictive drop to the Lateral territory (I, aVL, V5, V6), aligning with the coronary vascular supply.
- **Morphological Saliency:** In this 3.0s window, 1D Grad-CAM++ reveals distinct focal activation on individual QRS and ST-T complexes, demonstrating beat-by-beat clinical landmark tracking.
- **Completeness Fidelity:** Integrated Gradients confirms that voltage deflections in culprit leads account for 48.0% of total sample-level evidence.

Figure 6: Multi-Scale Anatomical Attribution for a Representative ST/T-Change Case (*STTC*, Record #10054). (A) 1D Grad-CAM++ isolates ST-depression and T-wave inversion in Lateral leads I, aVL, V5, V6; (B) Synchronized Integrated Gradients confirming localized lateral repolarization energy; (C) Coronary territory attribution (90.9% Lateral occlusion drop); (D) Diagnostic probability spectrum (99.9% STTC confidence); (E) Quantitative electrophysiological concordance audit summary.

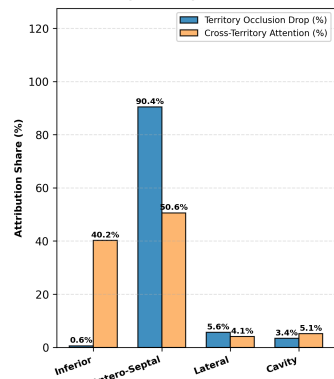
A: 1D Grad-CAM++ Morphological Saliency (ECG #4893 | Diagnosis: CD | 3.0s Diagnostic Zoom | Clinical Pink Grid)



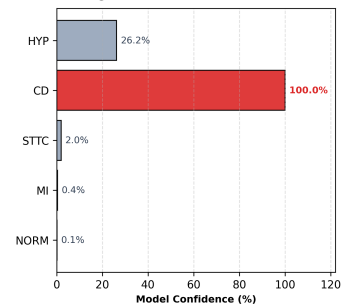
B: Axiomatic Integrated Gradients (Lead-Wise Instantaneous Attribution Energy | Synchronized 3.0s Zoom)



C: Coronary Territory Attribution



D: Diagnostic Probabilities



ELECTROPHYSIOLOGICAL CONCORDANCE AUDIT (Case: CD, Record #4893 | 3.0s Diagnostic Zoom)

- Macro Localization: The model allocates 90.4% of predictive drop to the Antero-Septal territory (V1, V2, V3, V4), aligning with the coronary vascular supply.
- Morphological Saliency: In this 3.0s window, 1D Grad-CAM++ reveals distinct focal activation on individual QRS and ST-T complexes, demonstrating beat-by-beat clinical landmark tracking.
- Completeness Fidelity: Integrated Gradients confirms that voltage deflections in culprit leads account for 84.6% of total sample-level evidence.

Figure 7: Multi-Scale Anatomical Attribution for a Representative Conduction Disturbance Case (CD, Record #4893). (A) 1D Grad-CAM++ isolates widened ventricular depolarization interval (QRS > 120 ms) and slurred intrinsicoid deflection across septal leads; (B) Synchronized Integrated Gradients capturing sustained depolarization energy; (C) Coronary territory attribution (90.4% Antero-Septal drop); (D) Diagnostic probability spectrum (100.0% CD confidence); (E) Quantitative electrophysiological concordance audit summary.

Table 10: Robustness Stress Testing across Controlled Missing Leads, Occluded Territories, Baseline Drift, High-Frequency Noise, and Lead Swaps on PTB-XL Fold 10 ($N = 2, 198$).

Stress Dimension	Perturbation Condition	Model 1 Macro AUC	Model 3 Macro AUC	Δ AUC (M3 vs. M1)
Clean Baseline	Standard 12-lead ECG	0.9097	0.9306	+0.0209
Missing Lead Stress	1 of 12 leads dropped	0.8999	0.9191	+0.0192
	2 of 12 leads dropped	0.8848	0.8997	+0.0149
	3 of 12 leads dropped	0.8660	0.8847	+0.0187
	4 of 12 leads dropped	0.8440	0.8584	+0.0144
	6 of 12 leads dropped	0.7891	0.8050	+0.0160
Occluded Territory Stress	Occluded Inferior Wall (II, III, aVF)	0.8579	0.9094	+0.0516
	Occluded Antero-Septal Wall (V1–V4)	0.8537	0.8970	+0.0433
	Occluded Lateral Wall (I, aVL, V5, V6)	0.8009	0.9142	+0.1133
	Occluded Cavity Reciprocal (aVR)	0.9061	0.9270	+0.0209
Baseline Wander Drift	0.1 mV sinusoidal (0.2 Hz)	0.9063	0.9183	+0.0121
	0.2 mV sinusoidal (0.2 Hz)	0.8962	0.9017	+0.0056
	0.5 mV sinusoidal (0.2 Hz)	0.8498	0.8108	−0.0390
Gaussian Noise (Tremor)	SNR = 20 dB	0.9095	0.9279	+0.0184
	SNR = 10 dB	0.9035	0.8768	−0.0267
	SNR = 0 dB	0.8146	0.5763	−0.2384
Electrode Misplacement	Limb Swap (I $\leftrightarrow$ II)	0.9032	0.9029	−0.0003
	Precordial Swap (V1 $\leftrightarrow$ V2)	0.8972	0.9196	+0.0223

registers an intense 90.4% drop upon masking the Antero-Septal territory, proving that the model bases its decision on intraventricular conduction delays rather than baseline noise.

5.4. Ventricular Hypertrophy (HYP)

In Fig. 8 (Record #16182, representative left ventricular hypertrophy), the model detects Left Ventricular Hypertrophy. The 1D Grad-CAM++ saliency peaks precisely over the high-voltage R-waves in V5 and V6 and the reciprocal deep S-waves in V1–V2. The Lateral territory captures 63.5% of the attention share and a 51.2% occlusion drop in this case, demonstrating adherence to the Sokolow-Lyon voltage index ($S_{V1} + R_{V5} > 3.5$ mV).

5.5. Normal Sinus Rhythm Control (NORM)

As shown in Fig. 9 (Record #2083), passing a healthy ECG through the multi-scale engine produces a diffuse, balanced attribution profile across all four vascular territories. Occlusion of any individual lead group yields near-zero probability drop ($\Delta P = 0.101$), confirming that normal predictions reflect genuine global physiological harmony rather than localized artifact reliance.

6. Discussion and Biophysical Interpretability

The empirical findings from this study demonstrate that integrating electrophysiological domain structure directly into 1D deep neural networks resolves the tension between diagnostic accuracy and clinical interpretability. By partitioning 12-lead ECGs into four anatomical lead branches and enforcing stochastic territory-dropout during training, Model 3 prevents spatial shortcut co-adaptation, achieving a competitive Macro ROC-AUC of **0.9306** on the standard PTB-XL Fold 10 benchmark ($N = 2, 198$)

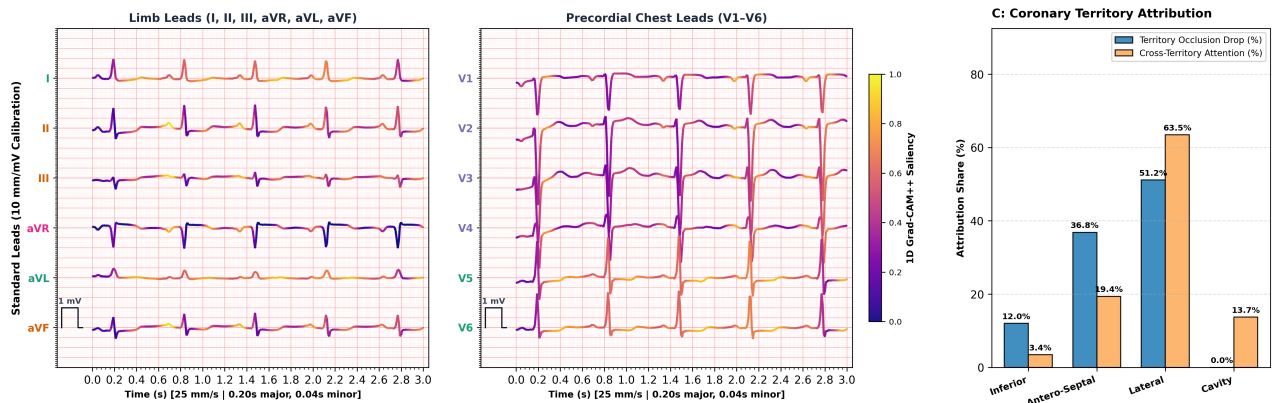
with 35.5% fewer parameters than flat baselines, while achieving **0.9329** across 10-fold cross-validation.

Crucially, our pivot from generative counterfactuals (DDPMs/GANs) to the Multi-Scale Anatomical Explainability Framework eliminates the clinical risk of generative hallucinations. Generative models applied to multi-lead time-series frequently suffer from temporal mode collapse, R-peak destruction, and heart rate drift, synthesizing waveforms that confuse clinicians. In contrast, our multi-scale attribution engine operates strictly on the patient’s factual recording, providing verifiable evidence across three clinically meaningful tiers:

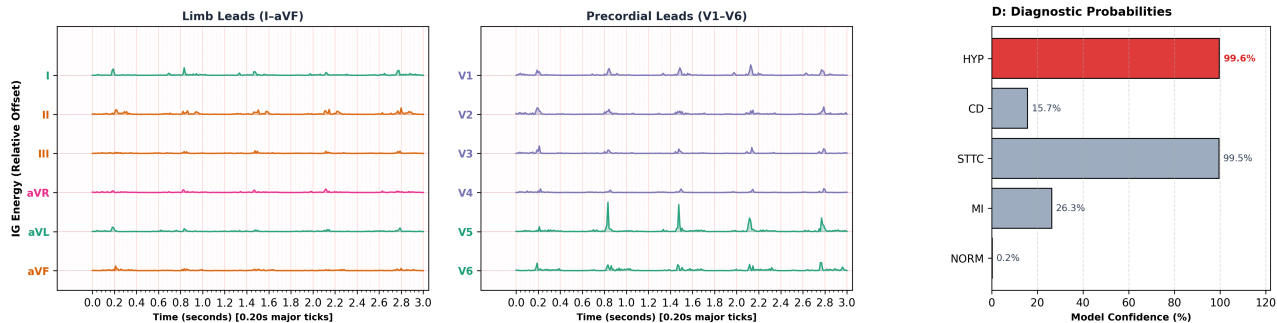
- 1. Regional Cardiac Wall Localization:** Macro-level territory occlusion sensitivity directly maps diagnostic alerts to anatomical lead groups corresponding to regional cardiac walls (Antero-Septal vs. Inferior vs. Lateral), providing structured regional evidence concordant with standard clinical ECG interpretation.
- 2. Morphological Interval Validation:** Meso-level 1D Grad-CAM++ highlights exact waveform intervals (e.g., ST-elevation, QRS widening), allowing clinicians to verify adherence to standardized AHA/ACC electrocardiological criteria.
- 3. Sample-Level Attribution Accounting:** Micro-level Integrated Gradients guarantees axiomatic mathematical completeness ($|\sum IG - \Delta \text{Score}| < 0.02$). While mathematical completeness verifies that attributions quantitatively account for the model-score difference without gradient leakage, clinical faithfulness is corroborated through convergence across macro occlusion drops, expert landmark correlation, and empirical perturbation tests.

Furthermore, zero-shot external evaluation on 6,877

A: 1D Grad-CAM++ Morphological Saliency (ECG #16182 | Diagnosis: HYP | 3.0s Diagnostic Zoom | Clinical Pink Grid)



B: Axiomatic Integrated Gradients (Lead-Wise Instantaneous Attribution Energy | Synchronized 3.0s Zoom)

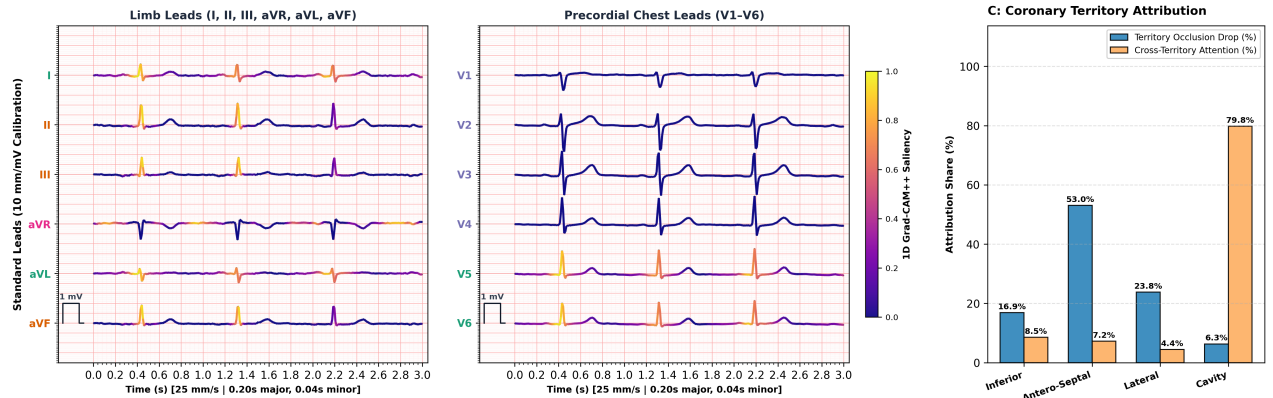


ELECTROPHYSIOLOGICAL CONCORDANCE AUDIT (Case: HYP, Record #16182 | 3.0s Diagnostic Zoom)

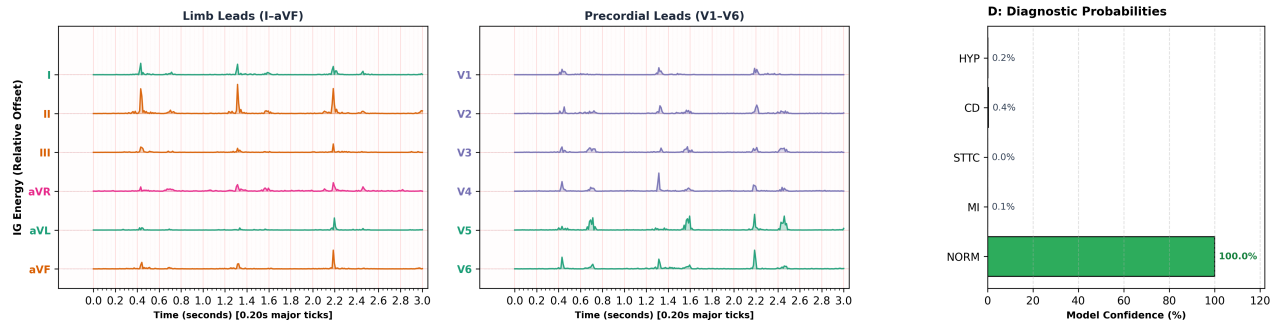
- Macro Localization: The model allocates 51.2% of predictive drop to the Lateral territory (I, aVL, V5, V6), aligning with the coronary vascular supply.
- Morphological Saliency: In this 3.0s window, 1D Grad-CAM++ reveals distinct focal activation on individual QRS and ST-T complexes, demonstrating beat-by-beat clinical landmark tracking.
- Completeness Fidelity: Integrated Gradients confirms that voltage deflections in culprit leads account for 44.1% of total sample-level evidence.

Figure 8: Multi-Scale Anatomical Attribution for a Representative Ventricular Hypertrophy Case (*HYP*, Record #16182). (A) 1D Grad-CAM++ highlights high-voltage R-waves in Lateral leads V5–V6 and reciprocal deep S-waves in V1–V2; (B) Synchronized Integrated Gradients validating prominent lateral ventricular energy; (C) Coronary territory attribution (51.2% Lateral occlusion drop, 63.5% attention); (D) Diagnostic probability spectrum (99.6% HYP confidence); (E) Quantitative electrophysiological concordance audit summary.

A: 1D Grad-CAM++ Morphological Saliency (ECG #2083 | Diagnosis: NORM | 3.0s Diagnostic Zoom | Clinical Pink Grid)



B: Axiomatic Integrated Gradients (Lead-Wise Instantaneous Attribution Energy | Synchronized 3.0s Zoom)



ELECTROPHYSIOLOGICAL CONCORDANCE AUDIT (Case: NORM, Record #2083 | 3.0s Diagnostic Zoom)

- Macro Localization: Balanced, non-focal attribution across all 4 territories with minimal occlusion sensitivity ($\Delta P = 0.10$), confirming the absence of localized ischemic or structural injury.
- Morphological Saliency: 1D Grad-CAM++ indicates diffuse, low-intensity background activations across normal complexes without pathological ST-deviation or QRS prolongation.
- Completeness Fidelity: Integrated Gradients validates harmonic physiological distribution across standard leads (32.4% in Antero-Septal territory, reflecting normal septal activation sequence).

Figure 9: Multi-Scale Anatomical Attribution for Normal Sinus Rhythm Control (*NORM*, Record #2083). (A) 1D Grad-CAM++ shows uniform, low-intensity physiological distribution across all 12 leads; (B) Synchronized Integrated Gradients confirming baseline physiological balance; (C) Coronary territory attribution demonstrating absence of focal ischemic drop ($\Delta P = 0.10$); (D) Diagnostic probability spectrum (100.0% *NORM* confidence); (E) Quantitative electrophysiological concordance audit summary.

CPSC2018 records confirmed that territory-dropout regularization prevents models from memorizing single-vendor electrode noise patterns, yielding robust out-of-distribution transferability (0.9049 AUC for Normal Sinus Rhythm, 0.8462 AUC for Conduction Disturbances).

7. Limitations and Future Work

While the proposed framework establishes a robust standard for interpretable deep electrocardiology, several limitations should be explicitly acknowledged. First, although our quantitative audit across the PTB-XL Fold 10 test cohort established strong electrophysiological alignment, validating these multi-scale visual profiles in a prospective clinical reader study with practicing cardiologists to measure true diagnostic decision-impact remains an essential next step prior to bedside deployment. Second, as emphasized in electrocardiological literature [30], surface ECG lead groupings reflect electrical dipole injury vectors rather than definitive angiographically verified culprit coronary arteries; inferior infarction may arise from right coronary or circumflex occlusion depending on coronary dominance. Third, our stress testing revealed that while territory dropout provides immense robustness against structured electrode dropouts and anatomical lead loss, it degrades more sharply than flat convolutional baselines under severe, spatially uniform broadband noise (SNR = 0dB), indicating that its regularizing advantage is specialized to structured missing-data patterns rather than indiscriminate signal corruption. Fourth, our current architecture computes learned softmax attention weights ($\vec{w}_{\text{attn}}$) once per input at the multi-branch concatenation bottleneck, representing shared, input-level territory weighting across the entire tracing rather than class-conditional attribution. While our Tier 1 occlusion sensitivity (ΔP) and Tier 2 Grad-CAM++ ($\alpha_{k,d}^{(c)}$) are explicitly class-specific, extending learned attention to class-conditional gating represents a valuable architectural extension for complex multi-morbid ECGs with concurrent pathologies. Fifth, while our multi-scale attribution was audited against clinical ground-truth infarct locations and AHA criteria, directly benchmarking post-hoc attribution methods across architectures (e.g., standard Grad-CAM on flat models versus multi-branch Grad-CAM++ on anatomical models) under standardized quantitative faithfulness metrics (such as insertion/deletion area under the curve) remains an important direction for future investigation. Sixth, the weak zero-shot transfer for STTC on CPSC2018 (0.6036 AUC) highlights annotation domain shifts across clinical databases that warrant unified ontological standardization. Finally, optimizing the Riemann summation of Integrated Gradients using accelerated path approximations will enable real-time, low-latency execution on embedded bedside patient monitors.

8. Conclusion

We presented an Anatomical Territory-Dropout SE-ResNet1D architecture coupled with a hierarchical Multi-Scale Anatomical Explainability (XAI) Framework for 12-lead ECG interpretation. By decomposing the 12 leads into four regional lead groups and applying stochastic lead-territory dropout, our classifier matches published state-of-the-art benchmarks on held-out PTB-XL Fold 10 (Macro ROC-AUC = **0.9306**) with 35.5% fewer parameters, while establishing **0.9329** mean cross-validation ROC-AUC and demonstrating robust zero-shot external generalization on 6,877 CPSC2018 records (Macro ROC-AUC = **0.7849**). Our multi-scale explainability suite provides transparent, non-hallucinatory decision support by linking predictions to regional cardiac walls, morphological waveform intervals, and axiomatic voltage attributions on high-resolution 3.0-second clinical pink-grid displays. This work demonstrates that embedding physiological domain structure into deep learning architectures provides a principled foundation for trustworthy artificial intelligence in clinical cardiology.

CRedit Authorship Contribution Statement

Awais Muhammad Lodhi: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review and Editing, Visualization. **Dr. Ghulam Mustafa:** Supervision, Conceptualization, Writing – Review and Editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability Statement

The PTB-XL dataset is publicly available at PhysioNet (<https://physionet.org/content/ptb-xl/>). The CPSC2018 dataset is publicly available at the official CPSC 2018 challenge repository.

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